CTA Strategy · Course Notes

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Contents

1 · What Is a CTA Strategy

1.1 What a CTA Trades

Definition 1.1 (CTA strategy)**.** A *CTA (Commodity Trading Advisor) strategy* is a rule-based strategy that trades **futures** systematically across asset classes—equities, fixed income, commodities, currencies—targeting **absolute return**, independent of market direction.

Note. The name is a historical artifact. Modern CTAs are not commodity-only; typical markets are equity-index futures, Treasuries, FX, energy, metals, and agriculture.

1.1.1 Index, ETF, or future

"S&P 500" names three different objects. Only two of them can be traded.

Layer	Example	What it is	Tradeable
Index	SPX, NDX	A published number computed from its constituents	No
ETF	SPY, VOO, QQQ	A fund holding the constituents; its own shares trade	Yes
Future	ES, NQ	A contract to settle at a future date; holds nothing	Yes

Definition 1.2 (Index)**.**

$$I_t = \frac{M_t}{D_t}, \qquad\qquad M_t = \sum_i N_i P_{it}$$

where P_{it} is the price of constituent i, N_i its float-adjusted share count, M_t the total float capitalisation, and $D_t > 0$ the *divisor*.

Claim 1.3. Index returns are the capitalisation-weighted average of constituent returns; the divisor sets only the level.

Proof. Both halves follow from I_t being a *sum divided by a constant.* A constituent's dollar move is N_i times its price change, so its share of the index move is its share of total capitalisation:

$$r_I = \sum_i w_{it} r_i, \qquad w_{it} = \frac{N_i P_{it}}{M_t}, \qquad \sum_i w_{it} = 1.$$

And D_t divides I_t at both dates, so it cancels from the ratio—it can shift the level anywhere without touching a single return. □

Note (Two consequences)**. The top names dominate.** At $w \approx 0.07$ against $w \approx 0.0002$, the largest constituent carries roughly **350 times** the index impact of the smallest; this is not an average of 500 *prices*. **And the level is arbitrary.** D_t cancels from returns, so the scale is whatever the base period set it to—for the S&P 500, 1941–43 = 10. **Compare returns, never levels.**

A price level is meaningful only against that same series' own history. Across tickers it carries no information at all—what each ETF's share price happens to be was fixed by how the fund was structured at inception. Dividing every series by its own first observation removes that arbitrariness:

Ticker	**Raw close, start**	**Raw close, end**	**Rebased**	**Return**
SPY	325.05	740.83	227.9	+127.9%
TLT	137.10	87.35	63.7	−36.3%
GLD	143.97	368.49	255.9	+155.9%

SPY has the highest closing price on every single day of the sample, yet GLD returned more. The end level alone cannot tell you this; you need the start it is measured against.

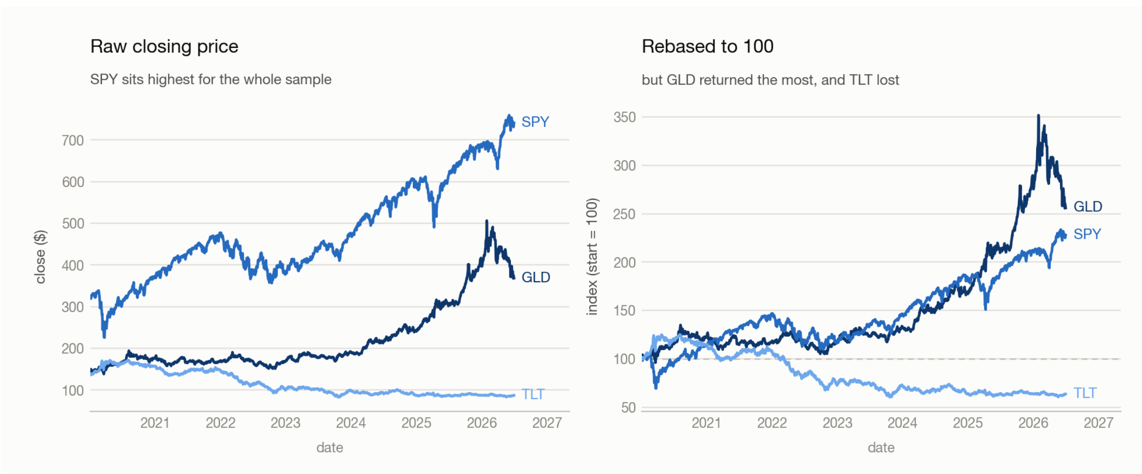


Figure 1.1: The same three ETFs, twice. Left, raw closing prices: SPY runs highest for the entire sample. Right, each series divided by its own first value and multiplied by 100: GLD now finishes highest and TLT ends below the line it started on. The ordering of the levels and the ordering of the returns are different orderings.

1.1.2 Why futures rather than ETFs

Reason	**Futures**	**ETFs**
Symmetry	Shorting costs nothing extra	Shorting needs borrowed shares, a fee, availability
Coverage	Deep markets in all four asset classes	Poor in commodities and FX
Capital efficiency	Margin is roughly 5 to 10 percent of notional; spare cash earns **collateral yield**	Full notional tied up

Symmetry is decisive: a strategy that must go short as freely as long cannot tolerate the asymmetry, and §1.3.3 shows why the short leg is the fragile one. The cost is that futures expire, so a continuous series has to be stitched together from contracts—see 100 · The Dataset.

1.2 Momentum, the Source of the Return

1.2.1 What momentum is

Definition 1.4 (Momentum). *Momentum* is the empirical tendency for recent relative performance to persist: assets that outperformed over the past one to twelve months tend to keep outperforming, and recent laggards tend to keep lagging.

Note. The claim is about *returns*, not price levels—it concerns the ordering of performance across assets and across time. That is why one momentum rule can be applied to markets with nothing else in common, from Treasury futures to natural gas.

Turning this into a number—a lookback window, a normalization, a test that it carries information—is 2 · Testing a Signal's subject. Here it is only the premise.

1.2.2 Why momentum might work

Two contested explanations, no proof. Momentum has been durably profitable regardless.

- **Information diffusion.** Unglamorous macro news spreads gradually, creating sustained pressure. Possibly weakened since 2008, as computing and data distribution got faster.
- **Selective momentum capture.** Winners keep winning and losers keep losing, because large positions take time to build and unwind, investors chase winners, and major events do not resolve in a single day.

2 · Testing a Signal turns this from a story into something testable.

1.3 Long and Short, the Two Sides of a Position

1.3.1 What the two sides mean

Definition 1.5 (Long)**.** Buy first, sell later. Profit is the sell price minus the buy price.

Definition 1.6 (Short)**.** Borrow and sell first, buy back and return later. Profit is the sell price minus the buy-back price.

Note. Running both deploys capital fully—a 150/50 or 200/100 book rather than idle cash—and lets the strategy express a negative view, which a long-only book cannot.

1.3.2 How a short is borrowed

A short is not selling out of thin air—the shares are borrowed before they are sold:

1. **Borrow** N shares from a lender—a broker, or a long-term holder.
2. **Sell** them now, and receive cash.
3. **Buy back** N shares later, from the market.
4. **Return** the N shares, closing the position.

Note. You never create shares. They are fungible, so returning "N shares of SPY"—not specific certificates—settles the debt. **Lenders are paid.** Long-term holders earn a **lending fee** on stock they were going to sit on anyway. **And the broker sits in the middle**, matching the two sides and holding margin; the market for this is **securities lending**.

→ The full mechanics beneath this—who holds the shares mid-loan, who receives the dividend, hard-to-borrow rates, and how a short squeeze runs away: §10 of the `market-101-foundations` note in `market_knowledge/`.

1.3.3 Why the short leg is the fragile one

Claim 1.7. For a static position, a long's loss is bounded by its outlay; a short's is not.

Proof. With signed share count s, entry price $P_0 > 0$ and exit price P_T, the payoff is

$$\mathrm{PnL}(P_T) = s\,(P_T - P_0)\,, \qquad P_T \in [0, \infty)$$

the domain being half-open because a price cannot go negative.

For a **long**, $s > 0$: PnL is increasing in P_T, so its infimum is at the left endpoint, $\mathrm{PnL} \geq s(0 - P_0) = -sP_0$—exactly the amount paid. For a **short**, $s < 0$: PnL is decreasing in P_T, and the domain has no right endpoint, so $\mathrm{PnL} \to -\infty$ as $P_T \to \infty$. □

The asymmetry is in the **domain**, not the payoff—both are lines of slope $|s|$. The floor exists only because prices are floored at zero.

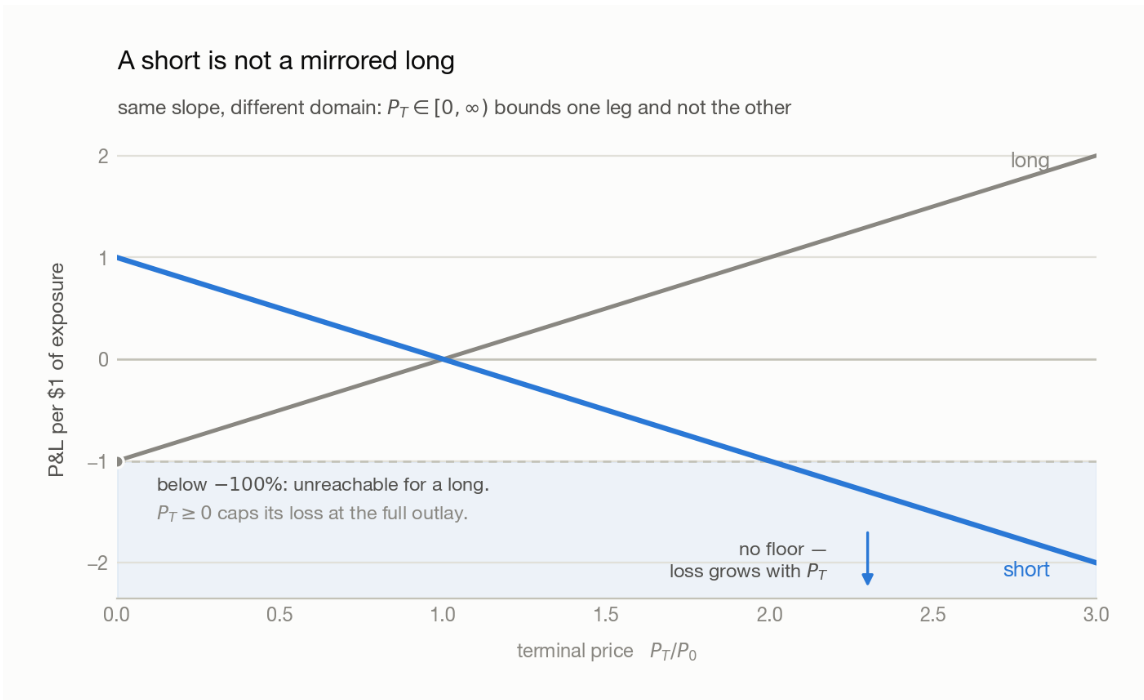


Figure 1.2: Profit and loss per dollar of exposure against the terminal price P_T. The long and short payoffs are lines of equal and opposite slope crossing at the entry price P_0; the long's line stops at −100 percent where the price reaches zero, while the short's continues downward without limit as the price rises. Illustrative.

Note. Hence margin requirements and forced buy-ins—and why a 150/50 book, 200 percent gross, is **two times leverage** rather than free money.

1.3.4 How a short is represented

A short needs no separate machinery. Carry the position as a **signed quantity** and let the sign do the work: a negative share count valued at the market price is a **negative position value**, which rises exactly when the price falls.

Note (Sign convention)**.** A negative position value is a **liability**—a debt owed—not a loss already taken.

Note (Static versus constant-exposure shorts)**.** The proof above holds the share count fixed at entry. A book that instead maintains a constant *dollar* exposure buys the short back down as it moves against it, so the unbounded loss is never realised. That daily rebalancing is an **implicit risk control**: the proof is right, but a constant-exposure book is not running the position it describes.

→ How this project implements it, with the accounting derivation and the measured gap between the two: `Backtests.md` in `Backtest_prototype/`.

1.4 Who Else Is Trading

Participant	**Examples**	**Typical hold**
Index / passive funds	Vanguard, BlackRock	Years to decades
Active managers	Fidelity, PIMCO	Monthly to quarterly
Hedge funds	—	Intraday to several days
Market makers	Optiver, Citadel Securities, IMC, SIG	Seconds to minutes
Noise traders	Retail, uninformed flow	No consistent horizon

Background

General finance knowledge needed to read this chapter, not part of its argument.

What is the relationship between futures/options and asset classes?

They answer different questions, so they are not alternatives to one another.

- **Asset class**—*what* underlying market you are exposed to.
- **Instrument**—*how* you hold that exposure: cash, futures, or options.

Futures are not an asset class, and options are not a rival one. Every asset class trades in every instrument:

Asset class	**Typical underlying**	**Futures**	**Options**
Equities	S&P 500, Nasdaq 100	ES, NQ	options on ES
Fixed income	US Treasuries, Bunds	ZN (10-year), ZB (30-year)	options on ZN
Commodities	Crude oil, gold, corn	CL, GC, ZC	options on CL
Currencies (FX)	EUR/USD, USD/JPY	6E, 6J	options on 6E

The same exposure comes in three wrappers: equity exposure is bought in cash as SPY, in futures as ES, or in options as an ES call. So in the opening definition—"trades **futures** systematically **across asset classes**"—the three parts are independent: **futures** is the wrapper, **across asset classes** is the breadth of underlying markets, and **systematically** means by rule or model rather than discretion.

Which asset class does a bond belong to?

Fixed income. Treasuries, corporate, municipal and high-yield bonds all sit there.

- “Fixed” describes the **contract**, meaning a scheduled coupon and the return of principal—not a fixed price.
- Bond prices still move, with interest rates, credit risk and time to maturity. That variation is what makes them a trend market at all.
- Treasury futures trade at 2-, 5-, 10- and 30-year maturities, so a CTA picks the *maturity* it wants exposure to.

Why do CTAs use futures rather than options?

	Futures	**Options**
Exposure	**Linear**—profit and loss move one-for-one with the underlying	Non-linear; depends on strike and moneyness
Extra drivers	None	Implied volatility, time decay, strike
What you are betting on	Direction	Direction **and** volatility

An option position is partly a volatility trade whether you intended it or not. Linear exposure is what lets one risk framework size 37 positions on the same scale.

Example 1.8. A trend follower might simultaneously hold long ES, short ZN, long GC, short 6J—all four are futures, and all four are different asset classes.

Appendix · Notation

Throughout, t is the date and i indexes an index’s constituents.

Symbol	**Means**	**First used**
I_t	the index level on date t—a published number, not a tradeable object	§1.1.1
P_{it}, N_i	constituent i’s price on date t, and its float-adjusted share count	§1.1.1
M_t, D_t	the total float capitalisation, and the divisor that turns it into the printed level	§1.1.1
w_{it}	constituent i’s share of total capitalisation—its weight in the index return	§1.1.1

Symbol	Means	First used
r_I, r_i	the index's return and constituent i's return, over the same period	§1.1.1
s	the **signed** share count of a position: positive is long, negative is short	§1.3.3
P_0, P_T	the entry and exit price of that position	§1.3.3
$\mathrm{PnL}(P_T)$	the position's profit and loss as a function of where the price ends up	§1.3.3
N	the number of shares borrowed to open a short	§1.3.2

Note (Collisions to watch)**.** s here is a **signed share count**, not 2's s, which indexes the asset. w_{it} is a **constituent's weight inside an index**, fixed by capitalisation and not chosen by anyone; 2's $w_{s,t}$ and 5 · §3.3's w_i are **portfolio** weights, which are decisions. r_i is a constituent's return, whereas elsewhere in the course r carries an asset and a date subscript and means the forward return being predicted.

2 · Testing a Signal

2.1 A signal is a hypothesis, not a formula

2.1.1 The three objects

Everything in this chapter is built from three numbers, one of each per asset per date.

Object	Symbol	What it is	Known at t?
Weight	$w_{s,t}$	the share of capital held in asset s on date t, signed—negative is a short	**yes**, you choose it
Signal	$MOM_{s,t}$	a number computed from data available at t, meant to say something about what comes next	**yes**, you compute it
Forward return	$r_{s,t}$	what the asset then goes on to deliver while the position is held	**no**—this is the unknown

Note (Scope)**.** This chapter describes **one asset at a time**. That is not a simplification: assets have different volatilities, so their raw signals are not on one scale and cannot be compared until §2.4 puts them there—and combining several into one book comes after that, in the position-construction step.

Definition 2.1 (Binary momentum)**.** The simplest member of the family—the sign of last period's return, and nothing else:

$$MOM_{s,t} = \text{sign}\left(r_{s,t-1}\right)$$

where s indexes the asset and t the date, $r_{s,t-1}$ is that asset's return over the period just ended, and $MOM_{s,t}$ is either $+1$ (long) or -1 (short), with nothing in between.

Trend direction survives; trend strength is thrown away. A window in which the asset rose 20 percent and one in which it rose 10 percent produce the *same* signal, so a strategy built on it takes the same size in both.

Definition 2.2 (Momentum)**.** Averaging over N periods rather than one, and keeping the value rather than its sign:

$$MOM_{s,t} = \text{Avg}\left(r_{s,t-i}\right) = \frac{1}{N}\sum_{i=1}^{N} r_{s,t-i}, \qquad i = 1 \ldots N$$

with N the lookback length and i the lag inside it—the average runs over the N returns ending yesterday, so at $N = 21$ it is last month's average daily move.

Example 2.3. Three assets over a five-day lookback:

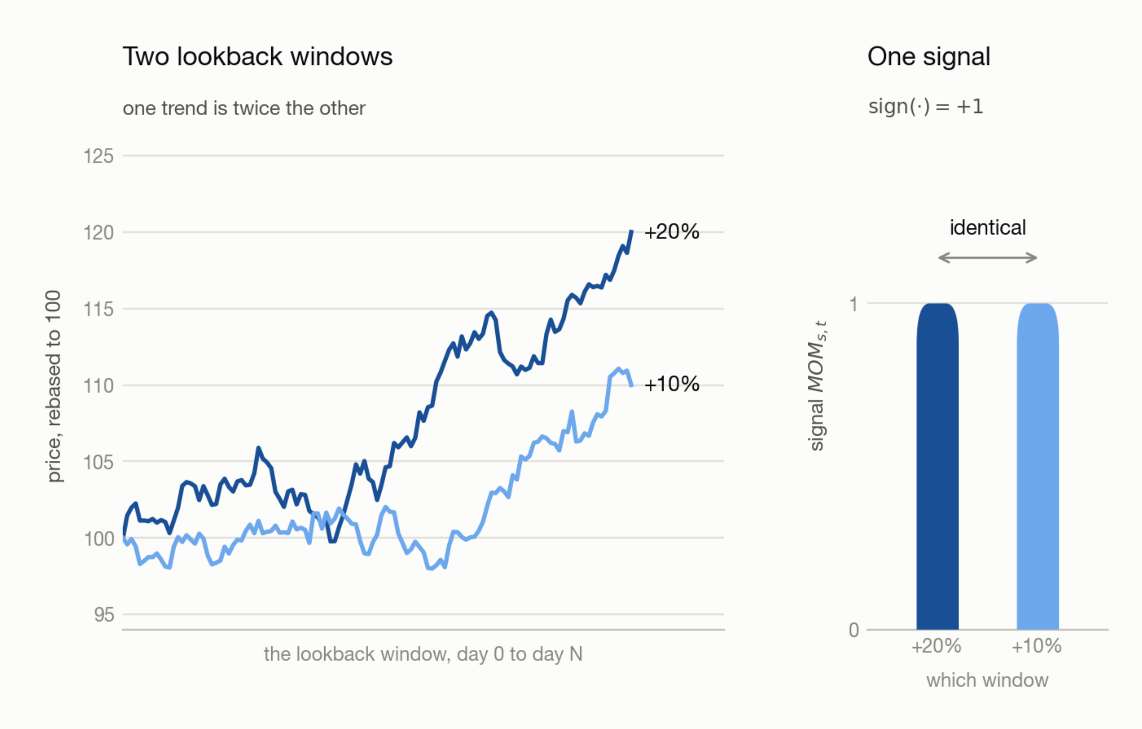


Figure 2.1: Left, one asset's price over two lookback windows, rebased to 100: one climbs to 120 and the other to 110. Right, the signal each path produces—two bars of identical height at +1. The magnitude of the move does not reach the signal. Illustrative.

Asset	**Last five returns**	**Avg → the signal**	**After** sign
A	+3%, +2%, +4%, +1%, +2%	**+0.024**	+1
B	+1%, −0.5%, +1%, 0%, +0.5%	**+0.004**	+1
C	−2%, −3%, −1%, −2%, −2%	**−0.020**	−1

The right column cannot tell A from B; the left makes A **six times** B, and since §2.1.2 sets each weight proportional to its signal, A is held six times larger. That ratio is the magnitude, and it is all the dropped sign was costing.

Note (The level is not yet meaningful)**.** 0.024 means nothing on its own—2.4 percent in a calm year and 2.4 percent in a violent one are not the same trend. Only *relative* sizes are ever used, and §2.4 is what puts them on one scale.

2.1.2 Why studying the signal is the same as studying the portfolio

A signal is not a strategy: it earns nothing and cannot be held. The reason to spend a chapter on one runs backwards, from what you are actually paid on—**so start from the portfolio, because that is what you optimize.**

$$R_t = \sum_s w_{s,t} r_{s,t}$$

R_t is the portfolio's return on date t, summed over every asset in the universe. Of the two factors in each term only one is yours—$r_{s,t}$ is the market's answer, and it arrives after the weight is already set. **So the whole of portfolio construction is the choice of $w_{s,t}$.**

But the weights are not free. Two sums over the book are fixed before any signal is consulted—the **net**, fixing how much *market* the book carries since longs and shorts cancel in it, and the **gross**, fixing how much *leverage* is deployed since they add:

$$\textbf{net:} \quad \sum_s w_{s,t} = 1 \qquad\qquad \textbf{gross:} \quad \sum_s |w_{s,t}| = G$$

with G the gross target—200 percent on a 150/50 book—and the net holding at 1 only to within a tolerance δ, since rebalancing is discrete. The two are independent: one asset at 100 percent and a book long 200 percent / short 100 percent carry the same net and three times the position, which is why gross is never free (1).

The weights are where the signal enters. You want weight where the forward return is about to be high, and that is precisely the number you do not have. The signal is the stand-in you put in its place, which means setting $w_{s,t} \propto MOM_{s,t}$.

Claim 2.4. Under that rule the portfolio's expected return is proportional to the correlation between the signal and the forward return, and to nothing else that can change its sign.

Proof. Take the signal and the return as centred. With $w_{s,t} \propto MOM_{s,t}$,

$$E[R_t] \propto \sum_s E\left[MOM_{s,t}\, r_{s,t}\right] = \sum_s \rho_s\, \sigma_{MOM,s}\, \sigma_{r,s}$$

using $E[xy] = \rho\, \sigma_x \sigma_y$ for centred x and y, where ρ_s is the correlation between asset s's signal and its forward return and $\sigma_{MOM,s}$, $\sigma_{r,s}$ their standard deviations. The dropped constant is a leverage choice and the volatilities are measurable properties of the asset—all three positive whatever the signal does. **Only ρ_s can carry a sign, so only ρ_s decides whether the portfolio makes money or loses it.** □

Note (The constraints do not disturb this)**.** Reaching a legal weight vector means **shifting** the signal until the net lands right and **scaling** until the gross does—both affine with a positive scale, under which correlation is unchanged: $\rho(a + bx, y) = \rho(x, y)$ for $b > 0$. The ρ measured on the raw signal is therefore the ρ governing the constrained portfolio, which is what lets the rest of the chapter ignore weights altogether; the position-construction step carries out both operations.

Everything collapses to that one number, and it is a number about the signal alone:

$$\textbf{hypothesis:} \quad MOM \uparrow \Longrightarrow \text{return} \uparrow \qquad\qquad \textbf{that is:} \quad \rho > 0$$

Write that down before the code—it names what would falsify the signal, and one you cannot falsify is a plot you will rationalize either way. Two things follow from it, and between them they set the shape of the rest of the chapter:

Consequence	Dealt with in
The portfolio never has to be built to test the signal. Measure ρ between the signal and the forward return directly, and you have measured the strategy	§2.2, §2.3
At equal ρ a volatile asset contributes more than a quiet one, since each term carries $\sigma_{MOM,s}\sigma_{r,s}$ alongside ρ_s. Ranking raw signals therefore ranks volatilities	§2.4

2.2 What you hoped for, and what you get

2.2.1 Which return the signal is scored on

Before signal and return can be plotted against each other, *the forward return* has to be pinned down: §2.1.2 measures ρ against it, but the phrase is not yet precise enough to compute. Three spans sit on the timeline, in this order:

Span	What it holds	Typical size
Lookback	the N returns the signal averages, ending at $t-1$	20–250 periods
Gap	g periods thrown away—neither averaged into the signal nor scored against it	0, a week, or a month
Paired return	$r_{s,t+g}$, the one period the signal is judged on	one period

The gap is the only free choice of the three—the -1 in **12-1 momentum** is exactly this gap, a twelve-month window stopping one month short of today.

Why leave one at all. Not executability: §2.1.1's window already ends at $t-1$, so $g = 0$ is knowable in time and anything tighter is look-ahead (§2.6). Executability fixes the floor at zero; **reversal** decides everything above it. A trend that has just formed hands a little of it back, and at daily frequency that rebate is large enough to cancel the edge outright—which is how a real signal arrives at the bar plot looking flat, or upside down. The mechanisms behind it are this chapter's Background.

Both directions cost something, which is what makes g a **hyperparameter** rather than a constant. Reversal strengthens as the sampling gets finer—the bid–ask bounce is a roughly fixed number of ticks per trade while the trend grows with the horizon—but a wider gap also throws away the opening of the trend. Set it from your trading frequency and a prior on how long the rebate lasts; running it at a day, a week and a month and keeping whichever bar plot looks best is what 5 · Modelling warns about.

Note (Neighbouring observations are near-copies)**.** Step one date forward and the lookback keeps all but one of its cells, as the figure's lower panel shows. Consecutive

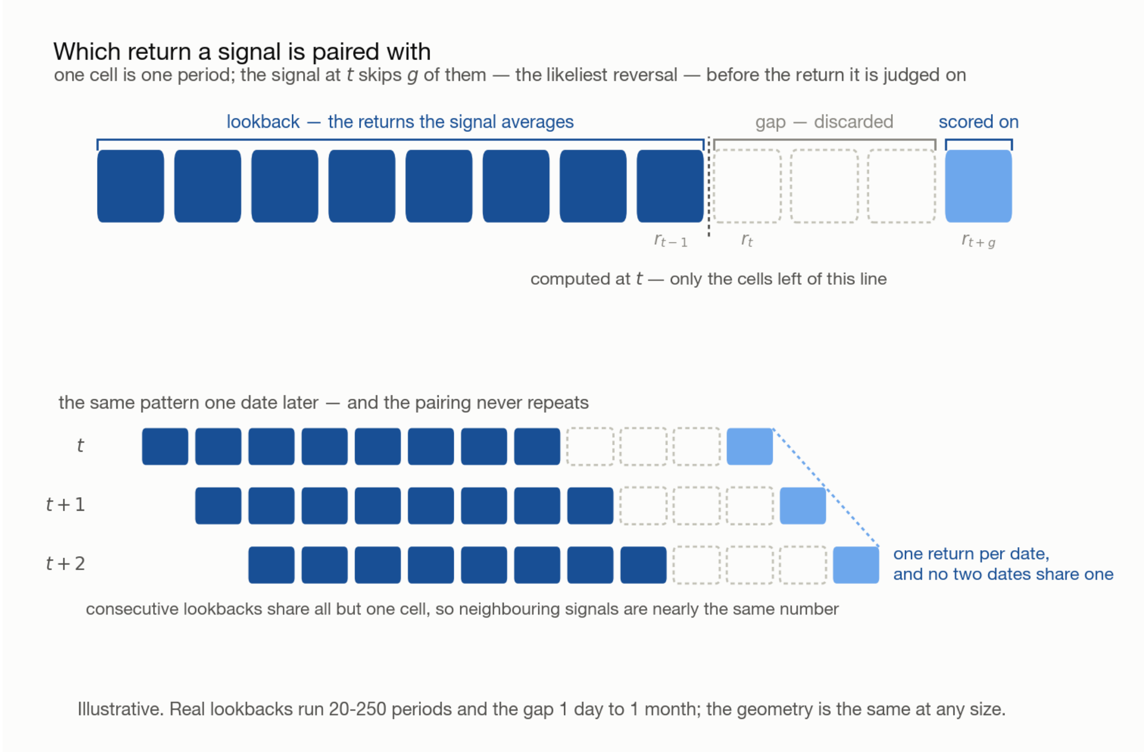


Figure 2.2: Above, one observation laid on a timeline of one cell per period: the lookback the signal averages, then the discarded gap g, then the single period the signal is scored on. Below, the same pattern for dates t, $t + 1$ and $t + 2$, each shifted one cell right—the lookbacks overlap in all but one cell, while the three scored cells form a descending diagonal, so no two dates share a scored return. Illustrative.

signals are therefore not independent draws, which is where §2.3.2.2's caveat on the effective sample size comes from.

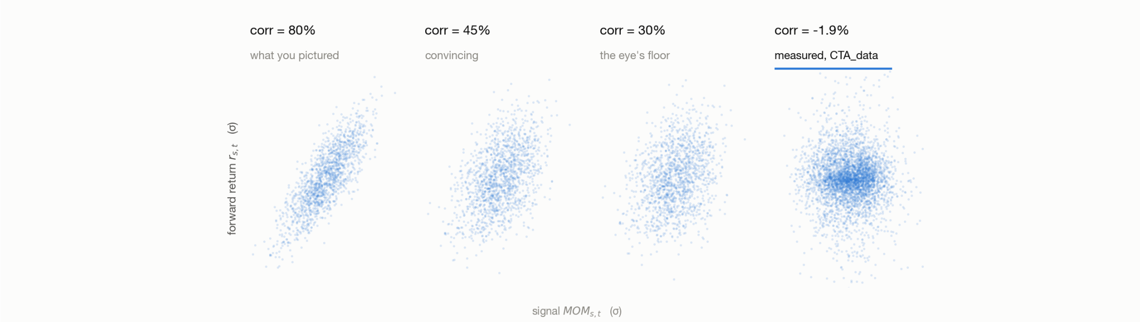
2.2.2 The plot everyone draws first

§2.1's hypothesis says forward return rises with the signal, so the first move is to plot one against the other. The picture you had in mind is the leftmost panel below. What comes back—for well over 99 percent of the scatters you will ever draw—is the rightmost.

2.2.3 Why the real one is a cloud

Claim 2.5. The scatter can neither confirm nor refute a signal, because the correlation a working signal carries sits far below the threshold at which the eye resolves a trend.

Proof. Both numbers are readable off the panels above: the eye stops resolving a tilt somewhere around 30 percent, and the fourth panel—21-day risk-adjusted momentum



First three panels drawn at a stated correlation. The fourth is measured on CTA_data/ — the correlation is the whole panel's, rendered from a 4,000-point subsample. Every panel holds a real relationship; only the first two show it.

Figure 2.3: Four scatters of forward return against a signal, both axes in standard deviations. The first three are drawn at correlations of 80, 45 and 30 percent from synthetic points; by 30 percent the tilt is barely arguable, which is the eye's floor. The fourth is measured on `CTA_data` at −1.9 percent—a formless cloud with a denser core, being real returns with fatter tails.

against the next session's return, measured over the 37 ETFs of 100 · The Dataset—comes in at **−1.9 percent**. Since $1.9\% < 30\%$, a signal that works and a signal that does not produce the same picture—**the absence of a visible trend is not evidence of anything.** □

So why is the correlation so small? Because the quantity being plotted is mostly not the signal.

Claim 2.6. Split the forward return into the part the signal reaches and the part it does not, $y = \beta x + \epsilon$. If ϵ is uncorrelated with x, the signal owns exactly ρ^2 of the return's variance, where ρ is their correlation.

Proof. Uncorrelated components add their variances,

$$\operatorname{Var}(y) = \beta^2 \operatorname{Var}(x) + \operatorname{Var}(\epsilon)$$

and on one regressor with an intercept the explained share is $R^2 = \rho^2$. At $\rho = -0.019$, $R^2 = 0.00035$—which does not mean "right 0.035 percent of the time", but: of the variation in forward return from one observation to the next, **0.035 percent is linear in the signal and 99.965 percent is not.** □

Example 2.7. Everything below is **one asset-date's** forward return, never a portfolio's, and every number is measured over the 57,649 asset-dates of the sample. The daily return standard deviation is $\sigma_y = 0.0149$—149 bp, a **basis point** being 0.01 percent:

Component	**Size**	**At** $\rho = -1.9\%$
Total return	σ_y	149 bp
What the signal moves	$\rho\sigma_y$	**2.8 bp**
What it does not	$\sigma_y(1-\rho^2)^{1/2}$	**149 bp**

The third row is not a rounding slip: at this correlation the signal removes so little that the unreachable part is the whole of it to three figures. Across an x-axis running from $-2\sigma_x$ to $+2\sigma_x$ the trend line moves about 11 bp end to end, while at every x the points scatter over roughly 596 bp. **A 0.11 percent slope inside a 6 percent cloud**—that ratio *is* the picture.

Note (Where a larger correlation would come from, and where it would not)**.** A predictor correlating 50 percent with next period's return would be arbitraged away long before you found it in anything liquid, so single digits is a competitive market's ceiling rather than a shortfall in craft. Two things do legitimately raise it and neither is on offer here: a **longer horizon**, since the signal's edge accumulates while the noise grows only as the square root of time, and **combining many signals**, since a composite reaches correlations no ingredient has on its own. Quoted numbers near 10 percent are almost always one of those two. A single signal against a single session is the hardest case in the family, and this chapter is deliberately standing in it.

2.3 Two ways out of the cloud

The scatter is unreadable, but the signal may still be in it. There are two things to try, and only the second one works.

2.3.1 Turn down `alpha`

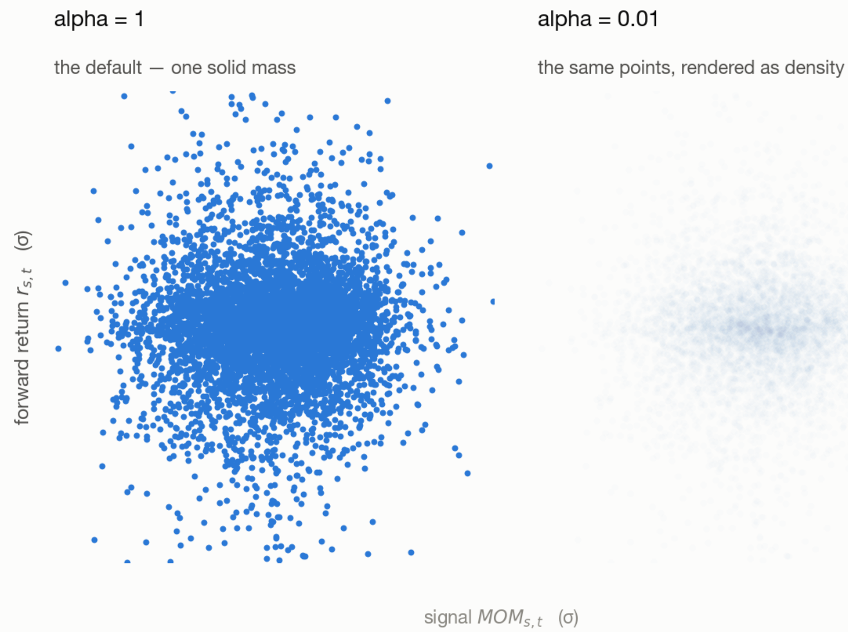
`alpha` is a marker's opacity. Turn it far enough down—`alpha=0.05` on a few thousand points, lower still on more—and a single point becomes nearly invisible, so only *overlap* renders and the chart is a density map rather than a mass of ink.

Note (Not the other alphas)**.** matplotlib's opacity keyword: not a regression intercept, not the excess return a manager is paid for, not 3's smoothing constant α. Code font is the tell.

Sometimes it is enough. A corner bitten out of the cloud—high signal never followed by a bad loss—would be a finding, because a signal that only rules something *out* is still tradeable. This data has no such corner, and that is the ordinary outcome.

Why it usually fails. Two reasons that compound:

- **The density comes back smooth.** Tens of thousands of asset-date cells average into a clean bivariate blob, and opacity renders that faithfully—a smooth density has no feature, and §2.2's band is fifty times taller than the tilt hiding in it.
- **It treats the symptom.** The scatter spends its resolution on individual noisy points when the claim is about their **average**; no rendering choice changes what is being rendered.



Measured on CTA_data/ — 6,000 asset-dates, both axes standardized. A smooth density renders faithfully as a smooth density.

Figure 2.4: The same measured asset-dates twice, both axes standardized. Left, at full opacity, they render as one solid disc with no internal structure. Right, the identical points at an opacity of 0.01: a smooth round density, denser in the core, with no thin region, no empty corner and no visible tilt anywhere.

Note (Look first anyway)**.** Always draw it. On the rare occasion something *is* readable—a curve, an empty corner—it beats every statistic, because it gives the *shape*. The mistake is concluding anything **from** a cloud.

2.3.2 Beta Method

Stop asking each point what it did, and ask each *group* of points what they did on average. The answer comes back as a **bar plot**, and the rest of §2.3 is about reading that one chart.

2.3.2.1 What a bar plot is

Definition 2.8 (Bar plot)**.** Five bars. The horizontal axis is the **signal**, coarsened into five ordered slots G1 . . . G5; the vertical axis is the **mean forward return** of the observations filed into each slot. Every observation lands in exactly one bar, and the error bar on it is the standard error of that mean.

Note (A bucket holds dates, not assets)**.** G5 is not a set of assets. It holds the **dates** on which this one asset's signal stood in the top fifth of its own history, and G1 the dates on which it stood in the bottom fifth. One asset passes through every bucket as the years go by, and the whole chart describes that asset alone.

Five steps build it:

1. **Fix one asset.** Its history gives one signal $MOM_{s,t}$ and one forward return $r_{s,t}$ per date.
2. **Rank the dates by signal value**, highest to lowest.
3. **File each date** into one of five slots by that rank: the highest fifth into G5, down to G1.
4. **Record** under each slot the forward return that date went on to deliver.
5. **Average** the returns filed under each slot.

Note (Step 2 is where this chapter's one defect lives)**.** Ranked on *what*, exactly? Take the raw momentum value and everything above still runs—but the asset's own volatility changes over the years, so +2 percent is a strong month in a calm tape and a quiet one in a violent tape, and step 3 files them into the same slot as if they were the same event. §2.4 is the correction, and until it lands the plot below should be read as machinery rather than as a result.

Note (What makes 57,649 dates from 37 assets one pool)**.** The measured row does not rank raw momentum across assets. Every signal entering it is already divided by that asset's own trailing volatility and ranked against that asset's own history before it lands in a bucket—the route §2.4 works out in full, with a worked two-asset example. G1 . . . G5 are therefore **percentile** slots, comparable across a Treasury ETF and a semiconductor ETF for the same reason they are comparable across 2021 and 2023: nothing but rank survives the trip, so pooling assets costs no more than pooling dates already did.

The sort key is the signal, never the return. Step 2 orders by $MOM_{s,t}$, which is known at t; step 3 only *records* what followed. G1 therefore holds the lowest-**signal** observations, not the worst performers.

Example 2.9. Five dates from one asset's history, invented, filed both ways:

Date	Signal	Forward return	Slot by signal	Slot by return
t_1	+2.1%	−0.4%	G4	G2
t_2	−1.8%	+0.9%	G1	G4
t_3	+0.6%	+1.3%	G3	G5
t_4	+3.4%	+0.2%	G5	G3
t_5	−0.9%	−1.1%	G2	G1

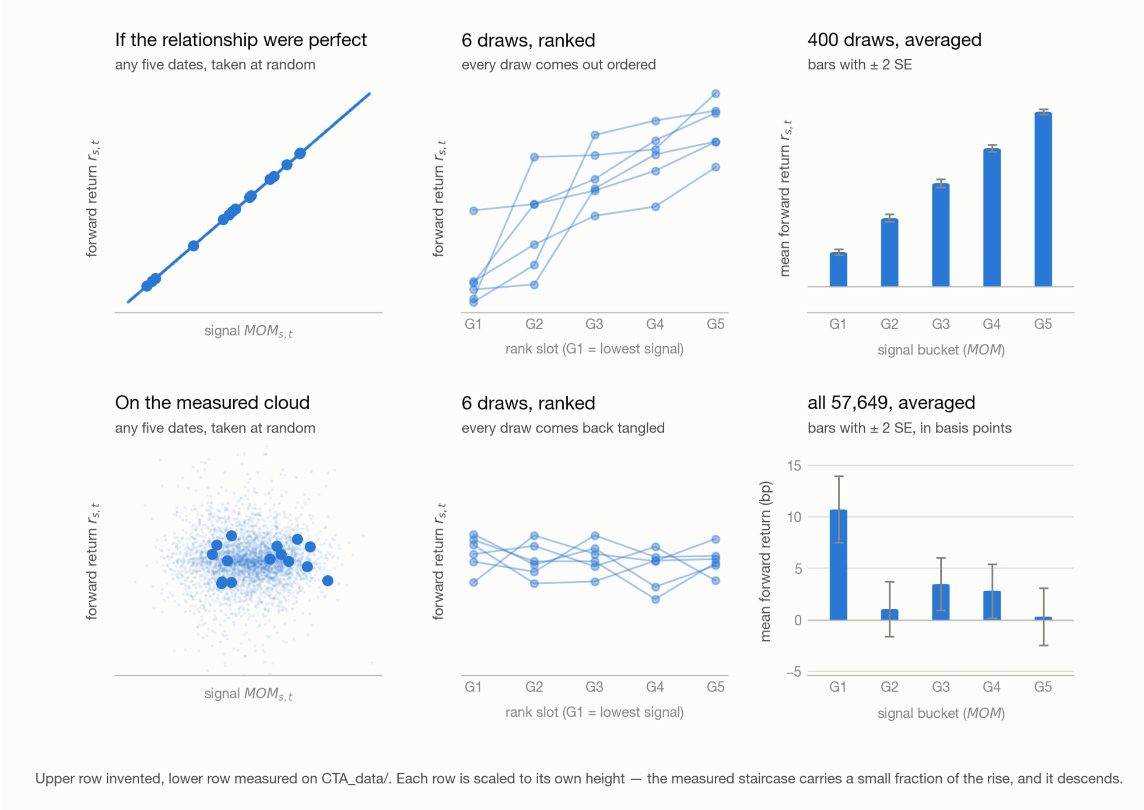


Figure 2.5: Six panels; the upper row is illustrative and the lower is measured on `CTA_data`. Left, the population of dates with fifteen drawn at random marked: on a perfect relationship they sit along a line, on the measured cloud anywhere. Middle, six draws of five dates plotted against the slots G1 to G5: on the line every draw rises monotonically, on the cloud the six tangle. Right, the answer with ±2 standard errors—a clean rising staircase above, and below the real chart, where G1 stands well clear of its error bar and G2 to G5 sit within a few basis points of zero, so the staircase descends.

Read down *slot by signal* and the returns come out unordered—which is exactly what five dates look like at a correlation this small. Read down *slot by return* and the ordering is perfect, and would be perfect for **any** signal including one straight out of a random number generator, because each bar is then reporting the sort key back to you. **The staircase is evidence only because the thing sorted on and the thing measured are different, and the second was not knowable when the first was computed.**

One bar therefore carries three pieces of information: which fifth of the signal's range it stands for, the average return that fifth went on to earn, and how much of that average is sampling noise.

2.3.2.2 How the bar plot shows a trend

The trend is the ordering, not the heights. §2.1's hypothesis was *higher signal, higher forward return*; on a bar plot that is precisely the statement G1 < G2 < G3 <

G4 < G5. A single bar's level moves with where the boundaries are cut (§2.4)—the ordering is what does not.

Two further readings carry information, and nothing else on the chart does:

Read	**What it establishes**
G5 − G1, measured against the error bars	the size of the edge next to what noise alone would draw. A rise that fits inside one error bar is not evidence of anything, and §2.3.2.3 turns that comparison into a test
The direction of the slope	a *descending* staircase is not a dead signal—it is the same edge carrying the opposite sign

A scrambled middle does not disqualify a signal: clean tails around a muddled G2–G4 is a common shape and a perfectly tradeable one, since the tails are where the positions go.

On direction, adopt a convention and keep it: **leave the signal's sign alone and let the chart come out upside down.** Flipping a signal the moment its staircase descends is cheap, and a dozen signals later you will not remember which ones you flipped. Read a descending staircase as *this works, traded short*, and carry the sign in the strategy rather than in the signal definition.

Claim 2.10. The ordering surfaces at all only because of step 4—averaging shrinks the noise and leaves the signal untouched.

Proof. Take m observations sharing a similar signal value. Their mean return still has expectation β times their mean signal—they were chosen for having nearly the same signal, so averaging them changes it barely at all—while the noise around it falls to $\sigma_\epsilon/m^{1/2}$. Measured, with the edge taken as the G5 − G1 gap the chart is trying to resolve:

	One observation	**Mean of 11,522**
Signal	10.4 bp	10.4 bp
Noise	149 bp	$149/11522^{1/2} \approx 1.4$ bp
Ratio	1 : 14	**7 : 1**

The left column is the scatter of §2.2 and the right column is one bar of the chart. Nothing was added—only the noise was taken away. □

The quantity being estimated is identical in all four panels—the real bucket means, +11 bp at G1 and roughly zero at G5. Only the error moves, and at $m = 30$ it is five times the whole staircase. **Sample size is not a detail of the recipe; it is the reason the recipe works.**

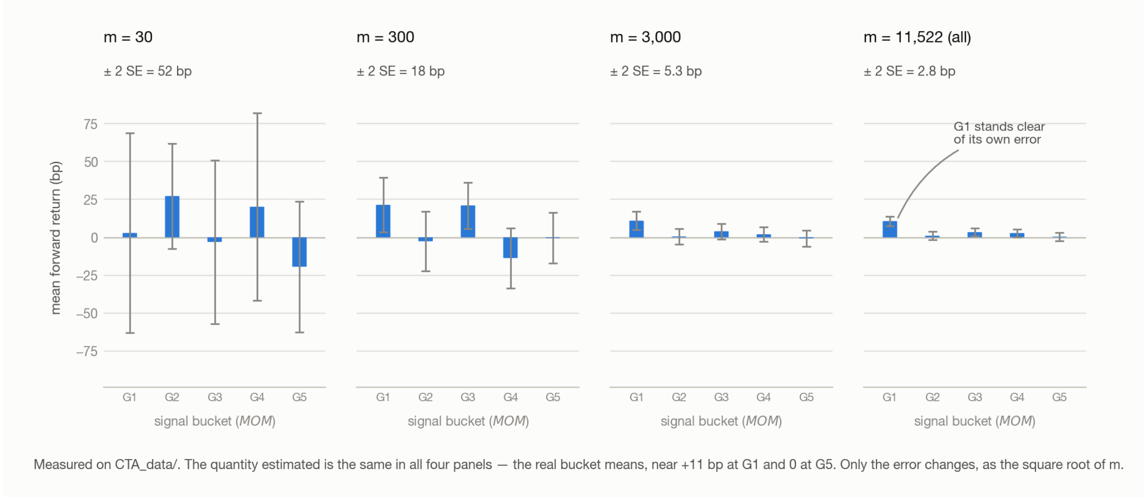


Figure 2.6: The same measured population as four bucket charts, mean forward return in basis points against signal bucket, drawn from 30, 300, 3,000 and all 11,522 observations per bucket. At $m = 30$ the bars swing over tens of basis points with error bars wider still and no ordering; by $m = 3{,}000$ G1 is visibly the tallest bar; at $m = 11{,}522$ the error bars are a few basis points and G1 stands clear of its own error while G2 to G5 sit near zero.

Note (Where that overstates it)**.** The $m^{1/2}$ assumes independence, and one asset's history violates it twice over: overlapping lookback windows share most of their inputs, and returns cluster in time, so consecutive observations are near-copies rather than independent draws. Both push the effective count well below m, and the noise does not really reach 1.4 bp. The logic survives: averaging cancels noise and leaves systematic signal.

2.3.2.3 Is the staircase real?

§2.3.2.2 read the rise against the error bars by eye. That comparison is a hypothesis test, and it is worth writing down, because the eyeball version and the arithmetic version do not agree.

Definition 2.11 (Standard error of a bucket)**.** For bucket g holding m_g observations whose forward returns have sample standard deviation σ_g,

$$\mathrm{SE}_g = \frac{\sigma_g}{m_g^{1/2}}$$

the spread of that bucket's **mean**, not of the observations inside it—which is why it shrinks with m_g while σ_g does not. Bars here are drawn at $\pm 2\,\mathrm{SE}_g$.

The hypothesis on trial is §2.1's, narrowed to the two buckets that carry the positions. A long book buys G5 and sells G1, so what has to be non-zero is the difference between those two means:

$$H_0 : \mu_{G5} = \mu_{G1}$$

against the alternative that they differ, where μ_g is bucket g's true mean forward return. The statistic is the observed difference in units of its own noise:

$$z = \frac{\mu_{G5} - \mu_{G1}}{\left(\mathrm{SE}_{G5}^2 + \mathrm{SE}_{G1}^2\right)^{1/2}}$$

the denominator adding **variances** rather than standard errors, because the two bucket means are computed from disjoint sets of dates.

Note (Why it is a t and gets called a z)**.** Both μ_g and σ_g are estimated from the same sample, so the exact null distribution is Student's t, not the normal. It stops mattering fast: at bucket sizes in the thousands the two are indistinguishable and the number is quoted as a **z-score**. Derive it as a t, report it as a z, and at m_g in the tens use the t and mean it.

Claim 2.12. $|z| \geq 2$ rejects H_0 at the 5 percent level.

Proof. Under H_0 the statistic is standard normal in the large-sample limit, and a standard normal places 0.0455 of its mass outside ±2—less than $\alpha = 0.05$, so the observation lies inside the two-sided rejection region. □

Example 2.13. The whole sample, five buckets, 11,522 dates in the smallest:

Bucket	μ_g	σ_g	m_g	SE_g
G1	**+10.72 bp**	173 bp	11,522	1.62 bp
G2	+1.05 bp	144 bp	11,523	1.34 bp
G3	+3.48 bp	136 bp	11,525	1.27 bp
G4	+2.83 bp	140 bp	11,523	1.30 bp
G5	**+0.31 bp**	149 bp	11,556	1.39 bp

$$z = \frac{0.31 - 10.72}{\left(1.39^2 + 1.62^2\right)^{1/2}} = \frac{-10.41}{2.13} \approx -4.9$$

Well past 2 in absolute value, and the bars—G1 spanning +7.5 to +14.0 bp, G5 spanning −2.5 to +3.1—clear each other with room to spare.

Note (Non-overlap is sufficient, not necessary)**.** The two readings differ, and in the direction that costs you signals. Bars of half-width 2 SE fail to touch when the difference exceeds $2\,(\mathrm{SE}_{G5} + \mathrm{SE}_{G1})$, while the test rejects when it exceeds $2\left(\mathrm{SE}_{G5}^2 + \mathrm{SE}_{G1}^2\right)^{1/2}$—and a sum is never smaller than the root of the sum of squares. On the measured numbers the eye demands 6.0 bp and the test demands 4.3 bp, so a difference between those two is one the picture calls inconclusive and the arithmetic calls significant. **Bars that clear each other are always significant; bars that touch may still be.** Use the chart to accept and compute z before rejecting.

The sign is negative, and that is a result rather than a failure. $z = -4.9$ says the two means differ, decisively; it says nothing about which way round they should be. What was measured is that **a high 21-day momentum is followed by a slightly worse next session than a low one** —the descending staircase §2.3.2.2 told you to expect and to keep. It is not a broken signal, it is **reversal**, and the Background section explains where it comes from and what removes it: put a gap between the signal's window and the return it is scored on, and the sign comes back.

Note (What the test does not license)**.** Two things sit outside it. The $m_g^{1/2}$ in every standard error assumes independent observations, and §2.3.2.2's closing note explains why one asset's overlapping windows are nothing of the kind—the effective count is well below m_g, so every z here is optimistic. And a threshold of 5 percent means one signal in twenty passes on noise, which is a fact about *one* test: running twenty variants and reporting the one that cleared 2 is not a test at all, and is 5 · Modelling's subject.

Failing to reject is a verdict on the measurement, not on the signal. $|z| < 2$ says the data did not separate the two means. Whether that is because they are equal, or because the error bars were too wide to tell them apart, is a question about m_g —and the two readings are not interchangeable. The smallest gap the test could have caught, at the four sample sizes of the figure above and a common σ_g of 149 bp:

m_g	SE_g	**Smallest detectable G5 − G1**	**What a null result there rules out**
30	27 bp	77 bp	Almost nothing. An edge large enough to build a career on is still consistent with it
300	8.6 bp	24 bp	Only edges several times larger than anything this chapter measures
3,000	2.7 bp	7.7 bp	Most of the range that matters, but not the part costs decide
11,522	1.4 bp	3.9 bp	The effect, down to a gap too small to survive execution anyway

Only the bottom row turns a null result into a finding. In the top row the gate has not been failed so much as attempted—**the honest report is "not measured yet", and the fix is more observations, not a different signal.**

2.3.2.4 What the bar plot cannot show

Every date contributes one observation—its signal's score, and the return that followed. Step 5 flattens all of them onto one picture, five slots wide.

Step 5 integrates over t, and an integral hands back an area, never the shape of the function underneath it. A staircase that is monotone over twenty years is equally consistent with an edge that held throughout and with one that worked for five years and was flat for fifteen.

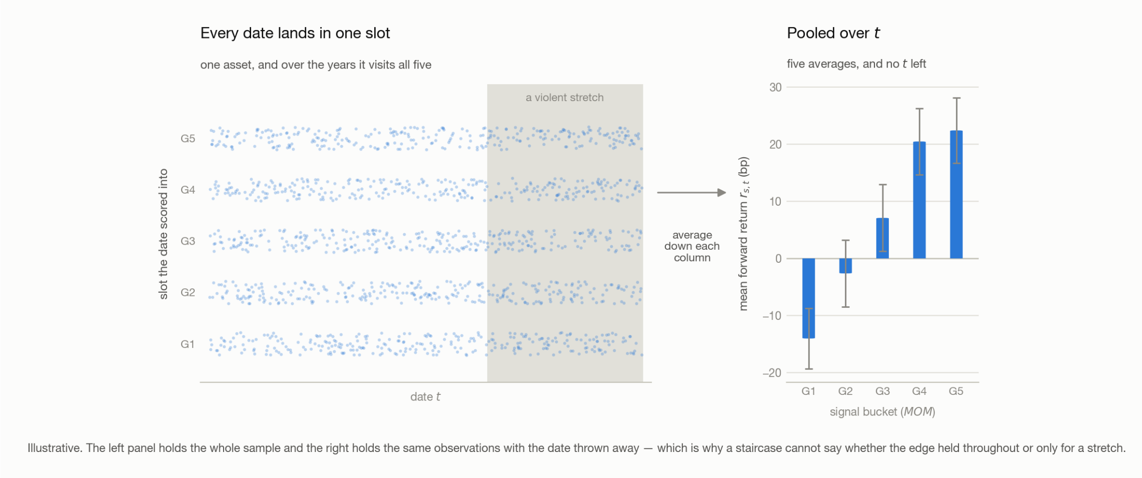


Figure 2.7: Left, one asset's dates strung along a time axis, each drawn at the slot its signal scored into, so all five rows are populated throughout and a violent stretch at the right is shaded. Right, the same observations with the date discarded: five bars of mean forward return with error bars. The arrow between them averages down each column, and the time axis is what it spends.

Not on the chart	**Where to look instead**
When the edge happened	the equity curve of 5 · Modelling
Who is inside a bucket—which dates, which regime	the dates behind each bar, tabulated rather than averaged
The spread of returns behind a bar, as against its mean	the distribution within a bucket, not its mean
How independent the observations are	overlapping windows, per the note above

2.4 Step 2, corrected — risk-adjusted momentum

This section repairs §2.3.2, it does not extend it. Every piece of machinery there survives: the sort key is knowable at t, the returns are recorded honestly, and the averaging shrinks noise exactly as the proof says. What §2.3.2 never settled is *what step 2 ranks*, and ranking the raw momentum value produces a chart about something other than the signal.

Raw momentum is not comparable **across time**, and §2.3.2 compares nothing else: its five slots sort one asset's own dates against one another. 2 percent in the calm 2021 tape and 2 percent in the 2023 rate-hike drawdown are different events, and filing them into the same slot says they are the same.

The identical failure reappears one step later, **across assets**, the moment you want more than one name in a book—and there it is an assumption you have to state rather than a fact. A universe of five hundred large-cap equities might plausibly sit on one scale already. A CTA universe does not: 50 bp in a session is an ordinary day for an equity index and, for a Treasury future, an event that would lead the news. 2 percent monthly momentum in a Treasury ETF is a large move; the same 2 percent in a semiconductor ETF is a quiet month. Rank the two against each other on raw momentum and the semiconductor wins every time—not because its trend is stronger but because everything about it is larger.

Divide by volatility:

$$MOM^{\text{risk-adj}}_{s,t} = \text{Avg}\left(\frac{r_{s,t-i}}{\sigma_{s,t}}\right), \qquad i = 1 \ldots N$$

where $\sigma_{s,t}$ is that asset's volatility, estimated on data strictly before t—a denominator that peeks at the future contaminates the signal as surely as a numerator would (§2.6).

Every date, and later every asset, now lands on one scale, so §2.3.2's slots compare like with like—two students both scoring 80 on different exams against different cohorts have not achieved the same thing.

Note (A second route to the same plane)**.** Dividing by $\sigma_{s,t}$ is not the only way. You can also replace the value outright by **its percentile against that asset's own past**, which lands every date on $[0, 1]$ by construction and needs no volatility estimate at all. Either route hands step 2 something it can legitimately rank.

Example 2.14. Two assets, each with five past signal values and one for today:

Asset	Its own past signals	Today	Beats	Percentile
A	−10, 8, 4, 0, −5	**+7**	four of five	**80**
B	−100, −80, 90, −60, 20	**−70**	two of five	**40**

Raw, +7 against −70 is not a comparison anybody should make. As percentiles, 80 against 40 is—and it says A is the stronger trend *relative to its own history*, which is the only sense in which the question has an answer. The cost is the one every rank charges: the magnitudes are gone.

Note (Which cut to use once it is scored)**.** The percentile route hands back a uniform score, so cutting at the quintiles gives five equal groups for free. The volatility route does not: a standardized signal is roughly normal, so cutting it at fixed intervals such as ±1 and ±2 leaves almost everything in the middle and starves the two tails you would actually trade. Cut it at its quintiles instead.

The rule this section leaves behind. Every signal is **standardized** or **scored against its own past** before it is compared to anything—otherwise the bar plot reports volatility while wearing the signal's name. The measured chart back in §2.3.2 already followed this rule before this section stated it: the percentile route is what let 37 assets share one pool of buckets.

2.5 What to report — and why it is not a Sharpe ratio

Every chart in §2.3 reports one quantity: **the mean forward return of a bucket, with its standard error.** The instinct at this point is to reach for something that sounds more like a result, and the one everybody reaches for is the wrong tool for this stage.

Definition 2.15 (Sharpe ratio)**.** A portfolio's average return in excess of what financing it costs, per unit of the volatility it ran:

$$\mathrm{SR} = \frac{E[R_p] - r_f}{\sigma_p}$$

with R_p the portfolio's return over a period, r_f the risk-free rate charged on the capital it used, and σ_p the standard deviation of R_p.

It is the right measurement for a finished portfolio and the wrong one here, for four separate reasons.

There is no portfolio yet. §2.1.2's argument is precisely that the signal can be judged *without* building one, and nothing in §2.3 has chosen a weight. A Sharpe ratio scores the output of a construction step that has not happened; computing one on a bucket's returns scores a portfolio nobody would run—equal weight on every date that fell into one fifth of the sample.

The financing rate is not zero here, and it is not constant. The numerator subtracts r_f, and which rate that is depends on the book's **net exposure** (§2.1.2):

Book	**Net exposure**	**Correct** r_f
Dollar-neutral long/short—one dollar long against one dollar short	≈ 0	**zero.** The short leg's proceeds fund the long leg, so no capital is being borrowed
A 150/50 CTA book	100 percent	**the actual financing rate**, on the capital the net exposure represents

Statistical-arbitrage books are the first row, which is where the habit of setting $r_f = 0$ comes from and where it is correct. A CTA book is the second row, and the rate belonging in it is a **time series**, not a constant: the policy rate this sample spans moved from near zero to above five percent. Carrying over "we always used 2 percent" from a book that was dollar-neutral puts an invented series into the numerator of every number reported.

It folds three estimates into one number. Mean, volatility and financing rate, each with its own error. A Sharpe that falls because the edge died and a Sharpe that falls because the tape got noisier are the same number—the exact opposite of §2.3.2's method, where one chart carries one quantity so a change has one cause.

A Sharpe means nothing without the market it was earned in. Two books, and the smaller number is the better strategy:

The tape	Market's own Sharpe	The book's Sharpe	Reading
Violent throughout	0.4	**0.8**	held together through amplitude that should have wrecked it, at twice the market's own risk-adjusted return—an excellent result
Calm and rising	0.8	**0.9**	barely beat holding the index. The extra 0.1 is within what one lucky bet would produce

The mean return is the quantity you can actually trade. It is what the book delivers, it is what §2.1.2's algebra is written in, and it is denominated in the same basis points as the costs that will be subtracted from it later. A Sharpe ratio is a score attached afterwards, useful for comparing **finished** portfolios against each other—which belongs to performance evaluation, after an optimization step has given it something to measure.

2.6 Information availability

Everything above assumes the signal at t uses only data knowable at t. Look-ahead bias is born here; the execution offsets of 5 · Modelling are the second line of defense and cannot rescue a signal contaminated at construction. The rolling-quantile rule (§2.4) and the train/validation/test split (5 · §1) are the same discipline.

Background

Why does a trend hand part of itself back, and can that be competed away?

Reversal is not an anomaly, it is the normal state of the tape. A trend that has just formed hands a little of it back, and at daily frequency that rebate is large enough to cancel the edge outright—which is why §2.2.1 leaves a gap g between the signal's window and the return it is scored on. Three mechanisms produce it, and they are not equally fragile:

Mechanism	What happens	Competed away?
Bid–ask bounce	Closes print alternately at the bid and the ask, oscillating around the true mid—negative serial correlation by construction	**No.** A measurement artefact; no price moved
Paying for liquidity	An order that must fill now pushes price past fair value, and whoever takes the other side is repaid when it returns	**No.** A fee for a service actually rendered
Overreaction	News is over-extrapolated, then partly corrected	**Yes**—the only part that decays

Two of the three are not mistakes, which is why reversal survives competition instead of being arbitraged into nothing—only the overreaction row decays as more people trade against it.

Appendix · Notation

Throughout, s indexes the asset and t the date. The signal is computed per asset, so s is dropped wherever a formula never leaves one asset; it carries weight everywhere the assets are ranked against each other.

Symbol	Means	First used
s, t	the asset, and the date in periods (days here)	§2.1
$r_{s,t}$, $MOM_{s,t}$, $w_{s,t}$	that asset's return in that period, the momentum signal it produces, and the share of capital that signal earns it	§2.1.1
R_t	the portfolio's return on that date, $\sum_s w_{s,t} r_{s,t}$	§2.1.2
G, δ	the gross-exposure target the weights must sum in absolute value to, and the tolerance allowed on the net	§2.1.2
ρ_s, $\sigma_{MOM,s}$, $\sigma_{r,s}$	one asset's signal-return correlation, and the standard deviations of its signal and its forward return	§2.1.2
N, i	lookback length, and the lag inside it running 1 to N	§2.1.1
x, y, β, ϵ	one observation's signal value and forward return, the slope between them, and the part of the return the signal cannot reach	§2.2.3
ρ, R^2	their correlation, and its square—the share of the return's variance the signal explains	§2.2.3
σ_x, σ_y, σ_ϵ	standard deviation of the signal, of the return, and of the unreachable part	§2.2.3

Symbol	**Means**	**First used**
m, m_g	observations sharing a bucket, and the count in bucket g	§2.3.2
μ_g, σ_g, SE_g	one bucket's true mean forward return, the sample standard deviation of the returns in it, and the standard error of its mean	§2.3.2.3
H_0, z, α	the null that two buckets share a mean, the standardized G5 − G1 difference testing it, and the level it is tested at	§2.3.2.3
G1 ... G5	the buckets, lowest to highest signal	§2.3.2
$\sigma_{s,t}$	one asset's volatility, estimated on data before that date	§2.4
R_p, r_f, σ_p	a portfolio's return over a period, the financing rate subtracted from it, and its return volatility	§2.5
g	gap length—periods between the end of the signal's window and the return it is scored on	§2.2.1

Note (Collisions to watch)**.** α is the test level here, and `alpha` in code font is matplotlib's opacity (§2.3.1); neither is 3's smoothing constant. σ carries whatever its subscript says: σ_g is the spread of returns *inside* a bucket, SE_g the spread of that bucket's mean, and $\sigma_{s,t}$ an asset's trailing volatility—not 5 · §4.2's σ_j, which rescales one signal. β is the slope of return on signal, never a market beta. g is the gap in §2.2.1 and a bucket index in §2.3.2.3; the subscript position tells them apart.

3 · Shaping the Lookback

2 · Testing a Signal left two blanks inside its own definition. Writing momentum as $\text{Avg}(r_{s,t-i})$, $i = 1 \ldots N$ fixes neither the length of the window nor the weight each return carries inside it—a flat average gives every return in the window an equal vote regardless of age, and commits to one window that cannot be both stable and timely.

An **exponentially weighted moving average (EWMA)** answers the first blank: replace the flat weights with a decay, so the newest return counts for the most and the average never quite forgets (§3.1). **Moving average convergence/divergence (MACD)** answers the second blank on top of that fix—instead of one EWMA it carries two, a fast one and a slow one, and trades the gap between them (§3.2, §3.3). MACD is therefore not a third technique standing beside EWMA; it is EWMA's own decay run twice, at two speeds, and read as a difference.

Both pieces run into the same wall no matter how they are tuned, and §3.4 is that wall: **a failure mode that survives every setting of either**, because it belongs to the market rather than to the parameters. The repair §3.4 can offer is local; the repair it cannot offer is 4 · Volatility Regimes.

3.1 EWMA instead of a simple moving average

3.1.1 Why the newest return should weigh most

A momentum signal is a weighted sum of past returns, and a simple moving average is one particular choice of those weights: make every one of them equal. Stated that way it is a claim about information—that a return from sixty days ago tells you exactly as much about where the asset stands today as yesterday's does.

It does not. **The newest return is the most recent evidence about the state the asset is in now**, so it should carry the most weight, and a return from three months ago the least. An exponentially weighted moving average is that preference written down.

Two things change at once. Relative to the box the newest lags are **over-weighted** and the old ones **under-weighted**—and the box's hard edge at twenty-one days is gone, so an EWMA thins out without ever quite forgetting.

3.1.2 The recursion, and the weights it produces

Definition 3.1 (EWMA)**.** Carry one running number and update it as each observation arrives:

$$\text{EWMA}_t = (1 - \lambda)\, a_t + \lambda\, \text{EWMA}_{t-1}$$

where a_t is the newest observation and $\lambda \in (0, 1)$ the decay—how much of the old average survives each step.

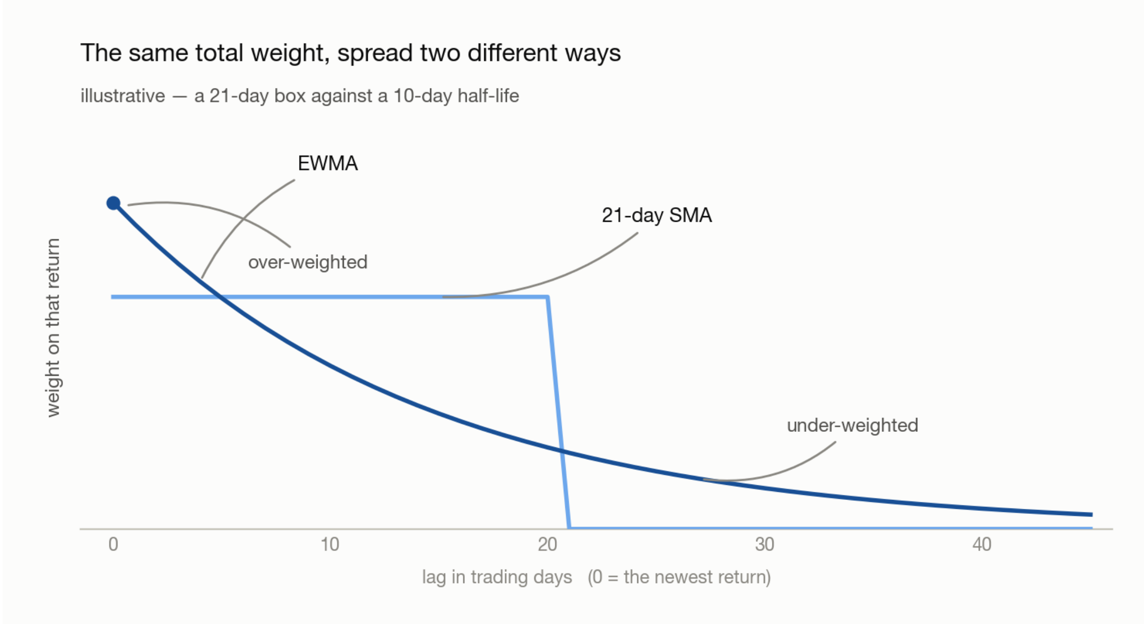


Figure 3.1: Weight carried by each past return against its lag in trading days, both curves normalised to the same total. The 21-day simple moving average is a flat box—identical weight out to twenty-one days, then nothing. The EWMA starts highest at lag zero, decays smoothly, and stays above zero past forty-five days. Illustrative.

Example 3.2. Take $\lambda = 1/2$ and three observations a_1, a_2, a_3, oldest first. Two updates run the whole thing:

$$\begin{aligned} \text{after } a_2 : \quad & \frac{a_1 + a_2}{2} \\ \text{after } a_3 : \quad & \frac{1}{2} \cdot \frac{a_1 + a_2}{2} + \frac{a_3}{2} \; = \; \frac{a_3}{2} + \frac{a_2}{4} + \frac{a_1}{4} \end{aligned}$$

Observation	**Age**	**Weight it ends up with**
a_3	newest	**1/2**
a_2	one step back	1/4
a_1	oldest	1/4

The newest observation carries half the total on its own, and each step back halves what is left—which is what *exponential* names. The oldest two tie only because the recursion had to start somewhere; run it long enough and that seam washes out.

3.1.3 Choosing how fast it decays

Tune the **half-life** H—the lag at which a return's weight has decayed to half the weight given to the newest one. Candidates are fractions of the lookback (1/2, 1/3, 1/5, 1/8), and—like every free parameter in this chapter—they are grid-searched rather than chosen; §3.2.2 grid-searches the analogous ratio for the fast/slow pairing §3.2 builds. In pandas the whole signal is `returns.ewm(halflife=H).mean()`.

No value of H is standard, and whatever a package ships as its default is an **empirical solution**—a number that fit somebody's historical data once. Every parameter in this chapter has that status, which is 5 · Modelling's subject rather than something to accept on authority.

3.2 Combining a fast and a slow horizon

§3.1 fixed the weight-shape blank with a decay. This section fixes the other one—not by finding a better single window, but by refusing to settle for just one.

3.2.1 Why two windows beat one

One lookback forces a choice between stable and timely. A long window rides a trend without being shaken out of it but arrives late at both ends; a short one turns on time, and turns on noise too. Rather than choose, carry both and trade **the gap between them**.

That gap is a single series—the fast average minus the slow one—and the rule is its sign: long while the fast average sits above the slow one, out when it drops back under. Two windows go in and one number comes out, which is what makes the pair an indicator rather than two rules run side by side.

What the crossing means. The fast average overtaking the slow one is not only an arithmetic consequence of the shorter window. It is the first evidence that flow has changed direction—that whoever was selling has stopped, or something large has started buying—while the move is still too small to register over a quarter. The same reading runs the other way and matters more there: the fast average rolling back under the slow one is what a large holder beginning to leave looks like from outside. Waiting for the slow window to turn on its own means selling to them on the way out.

3.2.2 Finding the pairing

The pairing is found by experiment, not chosen. Put the slow lookback N_s on one axis and the fast one N_f on the other, run 2's bar-plot test in every cell, and read the grid as a heat map.

Two things are true of that grid before any data is collected. **Half of it is empty by construction**—$N_f < N_s$ or there is no fast leg, so everything on and above the diagonal is struck out. And the candidates should be spaced **geometrically** rather than evenly: a day, a week, two weeks, a month, a quarter. Even spacing spends most of the grid on pairs that differ by a rounding error.

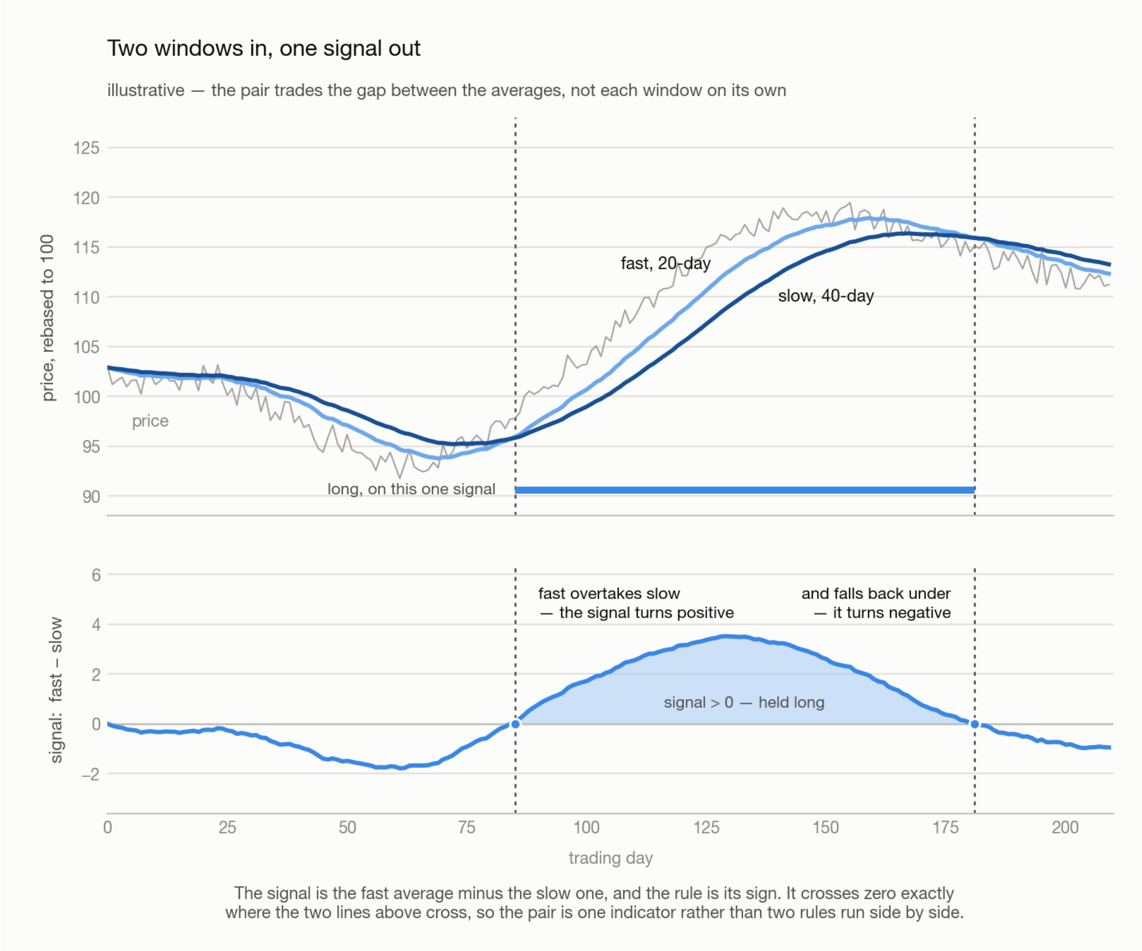


Figure 3.2: Above, one price path with a 20-day and a 40-day moving average over it; the fast average crosses above the slow one, and later drops back under, and the bar between those two dates marks the stretch the rule is long. Below, the single series the pair produces—fast minus slow—against zero, shaded where it is positive. Its two zero crossings fall exactly on the two dates the averages cross above. Illustrative.

What a diagonal band means. The warm cells run diagonally, and that shape is the finding rather than any single cell in it. On a geometric ladder a diagonal is a constant **ratio**, so what carries is N_s/N_f and not either window on its own—which is why the answer is quoted as a ratio at all. Research lands it near **2:1**, and MACD's 26/12 is one instance; for daily-to-weekly holding the band shows up around two weeks against a month. An empirical regularity, not a law, and 5 · Modelling's subject the moment you start trusting it.

Note (One hot cell is not a band)**.** A single dark square with pale neighbours is an artifact of however many parameter combinations were tried, not an edge. What earns belief is a contiguous warm region, because a real effect cannot switch off between one week and eight days.

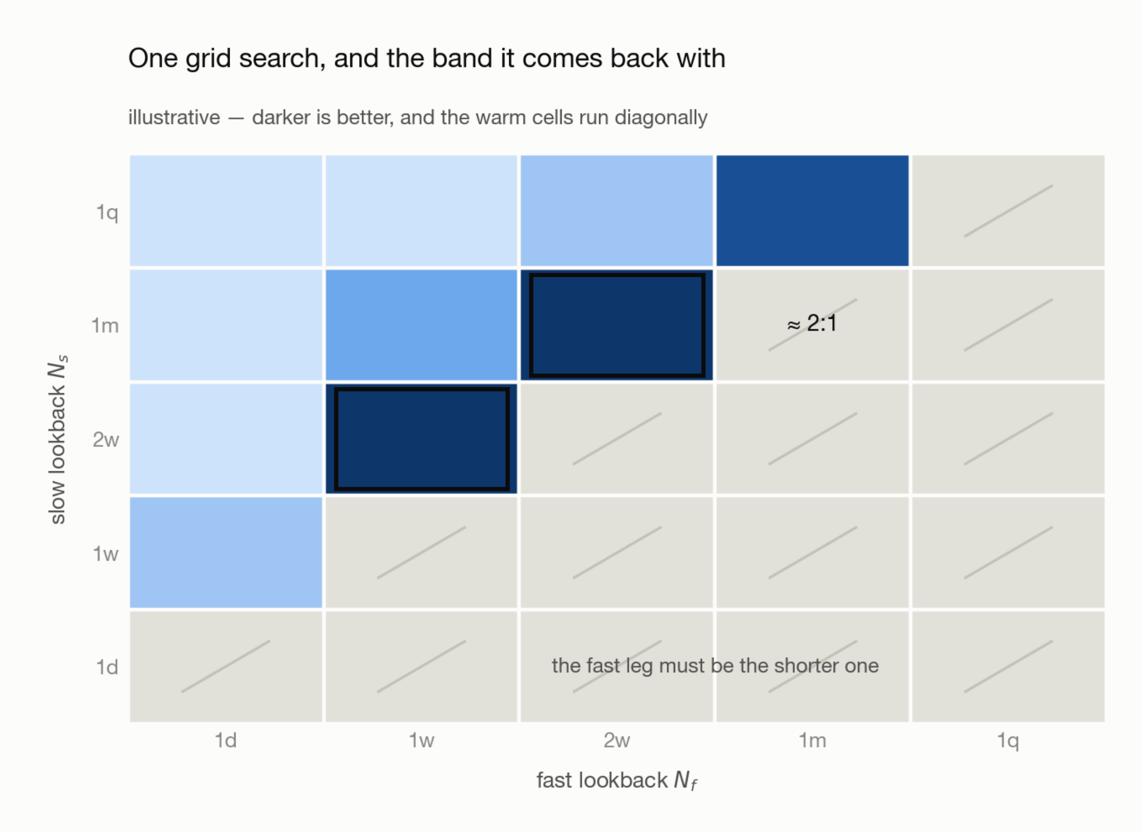


Figure 3.3: A five-by-five grid with a geometric ladder of lookbacks—one day, one week, two weeks, one month, one quarter—on both axes: fast across, slow up. Every cell on and above the diagonal is struck out, the fast leg having to be the shorter one. Of the rest, the darkest run in a diagonal band, and the two annotated cells sit at roughly two to one. Illustrative.

3.2.3 Where the combination pays

Best in **commodities**, and the reason is what moves them. An equity price answers to earnings, to the firm's own performance, to whoever is running it—many small idiosyncratic forces, none of which trend for long. A commodity answers to global energy supply, shipping, and whether the large participants are in the market at all, and it is far too large for one fund to push on its own, so a move that starts tends to be everyone moving together. Equities second. **Bonds** weakest: lean on one hard enough and someone takes the other side and presses it back.

3.3 MACD, stated precisely

§3.1 gave a window a decay; §3.2 took the difference of a fast average and a slow one and read its sign. That difference already has a name, and two further series conventionally travel with it.

3.3.1 The three series

Written out, MACD is three series built from two exponential moving averages of the **price**.

Definition 3.3 (MACD)**.** For a fast and a slow span $n_f < n_s$, each with its own smoothing constant $\alpha = 2/(\text{span}+1)$:

$$\begin{aligned} \text{MACD}_t &= \text{EMA}_{n_f}(P)_t - \text{EMA}_{n_s}(P)_t \\ \text{signal}_t &= \text{EMA}_9(\text{MACD})_t \\ \text{hist}_t &= \text{MACD}_t - \text{signal}_t \end{aligned}$$

where P_t is the price on date t and $\text{EMA}_n(P)_t$ its exponential moving average over span n. The asset subscript is dropped throughout—MACD is computed one asset at a time. Conventionally $n_f = 12$, $n_s = 26$, and 9 for the signal line.

Series	**Reads as**	**Sign means**
MACD line	recent average price versus a longer one	trend direction
Signal line	a smoothed MACD	the level MACD is crossing
Histogram	MACD minus its own smoothing	trend acceleration

3.3.2 Why it is momentum, not a separate indicator

Claim 3.4. MACD is a momentum signal. It is a weighted sum of past returns, differing from a lookback mean only in the shape of the weights.

Proof. Each EMA is a weighted average of past prices whose weights sum to one, so their difference weights the price i days ago by c_i, and those weights sum to **zero**:

$$\text{MACD}_t = \sum_i c_i P_{t-i}, \qquad c_i = \alpha_f(1-\alpha_f)^i - \alpha_s(1-\alpha_s)^i, \qquad \sum_i c_i = 0.$$

A zero-sum weighting is unchanged when every price shifts by a constant, so subtract P_t from each term and write $P_t - P_{t-i}$ as a sum of one-period changes $\Delta_{t-j} = P_{t-j} - P_{t-j-1}$. Collecting the coefficient of the change at lag j gives the **kernel** k_j:

$$\text{MACD}_t = \sum_j k_j \Delta_{t-j}, \qquad k_j = \sum_{i \le j} c_i = (1-\alpha_s)^{j+1} - (1-\alpha_f)^{j+1}.$$

Since $\alpha_f > \alpha_s$, every $k_j \ge 0$: MACD is a non-negative weighted sum of past price changes, exactly like a lookback mean. □

Note (What actually differs)**.** The kernel. A 21-day momentum weights the last 21 returns equally and everything older at zero; MACD's weights rise from a small value at lag 0, peak around lag 8, and decay without ever reaching zero. It therefore discounts *yesterday* relative to last week—deliberately, since the newest return is the noisiest—and never fully forgets.

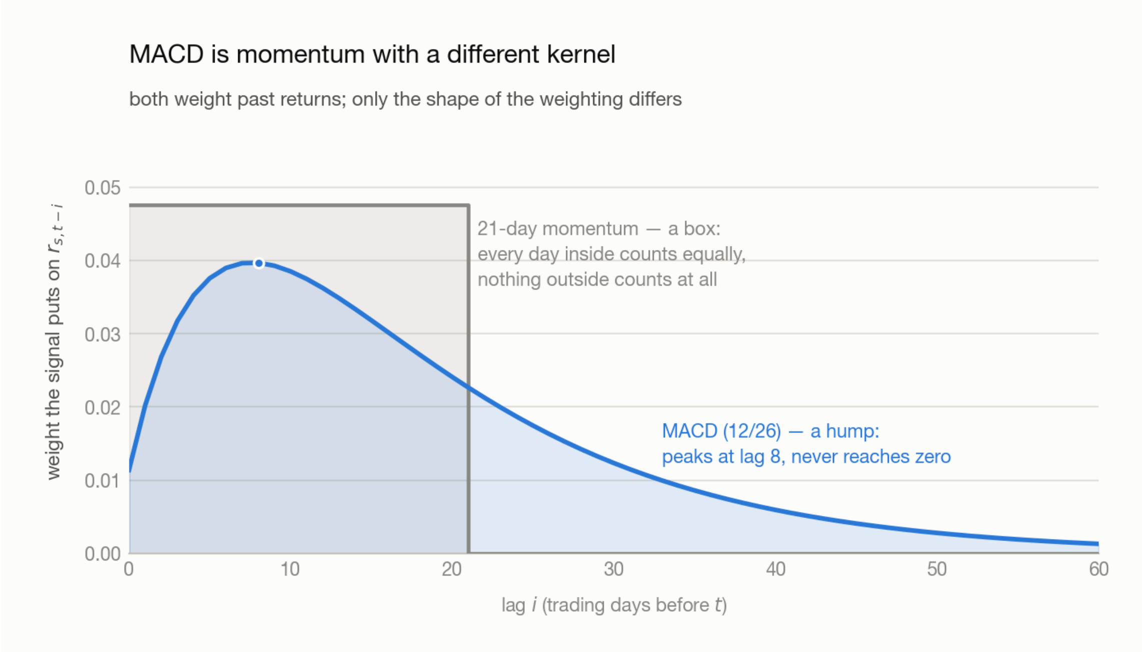


Figure 3.4: Weight given to each past daily return against its lag in trading days, for two signals normalised to the same total. A 21-day momentum is a flat box: equal weight for the last 21 days and zero beyond. MACD with spans 12 and 26 is a hump that starts low at lag zero, peaks around lag 8, then decays slowly and never reaches zero within sixty days. Illustrative.

Note (A choice, not a theorem)**.** §3.1.1 argued that the newest return deserves the most weight, and the most widely used trend indicator in the market does the opposite. Both positions are defensible: the newest return is the most recent evidence about the state the asset is in, and it is also the one carrying the most reversal (2's Background). Which wins is an empirical question settled by the bar plot, not a matter of taste—which is why §3.1.1 was written as a preference rather than a result.

3.3.3 From a value to a rule

Three common rules, in increasing order of information kept. All three still have to pass §2.3.2's bar plot before they earn a backtest.

Rule	**Go long when**	**Costs**
Zero-line	MACD is above zero	a slow trend filter, late
Crossover	the histogram is above zero	earlier, noisier—the churn is §3.4's subject
Proportional	always, sized by the standardized value	none of the magnitude, but see §2.4

The three differ in how much of the signal's magnitude they keep. They also differ in how *often* they trade, and that second axis is not a property of the rule alone: a crossover flips whenever the two legs touch, and how often they touch is set by how large the market's moves happen to be this month. Every parameter above is now chosen, and that count is still not under their control.

3.4 Volatility clustering breaks both

Definition 3.5 (Volatility clustering)**.** Large moves are followed by large moves and small by small, **regardless of sign**. Returns themselves are close to uncorrelated from one day to the next, but their *absolute* values are strongly and persistently autocorrelated. Direction is unpredictable; **amplitude is not**—a market has stretches, days to months, where everything is simply bigger, without the underlying trend having changed at all.

Neither piece survives it. A short EWMA half-life and MACD's fast leg are the same object under two names—a short average of recent returns—and a short average's swings scale with the amplitude around it. Enter a cluster and the sign starts flipping repeatedly, not because the trend turned but because the noise grew.

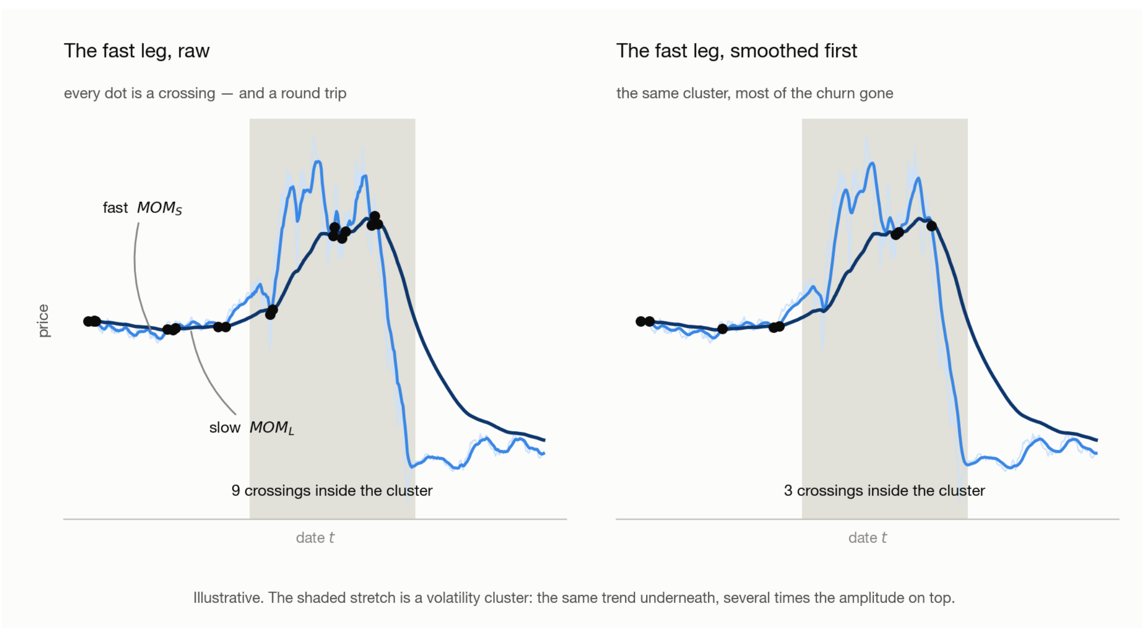


Figure 3.5: One price path, calm at both ends with a shaded volatility cluster in the middle, carrying a slow moving average and a fast one, every crossing marked. Left, the raw fast leg crosses the slow leg nine times inside the cluster. Right, the same fast leg passed through a short smoother first crosses three times. Illustrative.

The crossings are not merely expensive, they are backwards. A downward spike drags the fast leg under the slow one and flips the book short—at the bottom of the spike. The snap-back flips it long again, at the top. Repeat that through a cluster and the rule has been buying high and selling low on a schedule set by noise, which is worse than holding nothing: **whipsaw is not scatter around the right answer, it is the opposite of the right answer**, taken over and over. Nine crossings instead of three is nine round trips instead of three, and they land in exactly the stretch where spreads are widest and depth is thinnest—at forty crossings a year and five basis points a round trip, that is 200 bp of annual drag against a gross edge that might be four hundred.

The local fix is a shorter second average, or a margin. A **smoother**—a second average applied to the signal after it is computed, shorter than the signal's own period or it becomes another slow leg—stops a single day's noise from moving the sign on its own; MACD's 9-day signal line is one (§3.3.1). A **deadband**, which requires a crossing to clear some margin before the position flips rather than acting on the exact touch, is the other. Both read the signal and nothing else, so both are local: neither knows the tape has turned violent, each damps every date alike, and both pay on the quiet days to protect the loud ones. Going further means measuring the state of the market itself and asking what the rule is worth in each of its values.

→ 4 · Volatility Regimes.

Appendix · Notation

Throughout, t is the date. Everything in this chapter is computed one asset at a time, so the asset subscript s of 2 is dropped.

Symbol	**Means**	**First used**
a_t, λ	the newest observation, and the EWMA decay that keeps λ of the old average	§3.1.2
H	EWMA half-life, in periods	§3.1.3
N_f, N_s	fast and slow lookback lengths, in periods	§3.2.2
P_t, Δ_{t-j}	price on that date, and the one-period change at that lag	§3.3.1
n_f, n_s	fast and slow EMA spans, conventionally 12 and 26	§3.3.1
α_f, α_s	their smoothing constants, two over span plus one	§3.3.1
c_i, k_j	MACD's net weight on the price at that lag, and its kernel weight on the price change	§3.3.2

Note (Collisions to watch). N_f, N_s are plain lookback lengths and n_f, n_s are the EMA spans that play the same two roles—case is the tell. k_j is a kernel weight, not a portfolio weight: 2 reserves $w_{s,t}$ for the latter, which is why the kernel is not written w. And α here is a smoothing constant, not §2.3.1's `alpha` plotting keyword and not a regression intercept; code font against maths is the tell.

4 · Volatility Regimes

Three steps, in order. **MACD stops working when the tape gets loud** (§4.1). **The option market prices that loudness before any trailing estimate can measure it** (§4.2). **The resulting label is used by conditioning on it, not by adding it to the signal** (§4.3).

4.1 Why MACD breaks when the tape gets loud

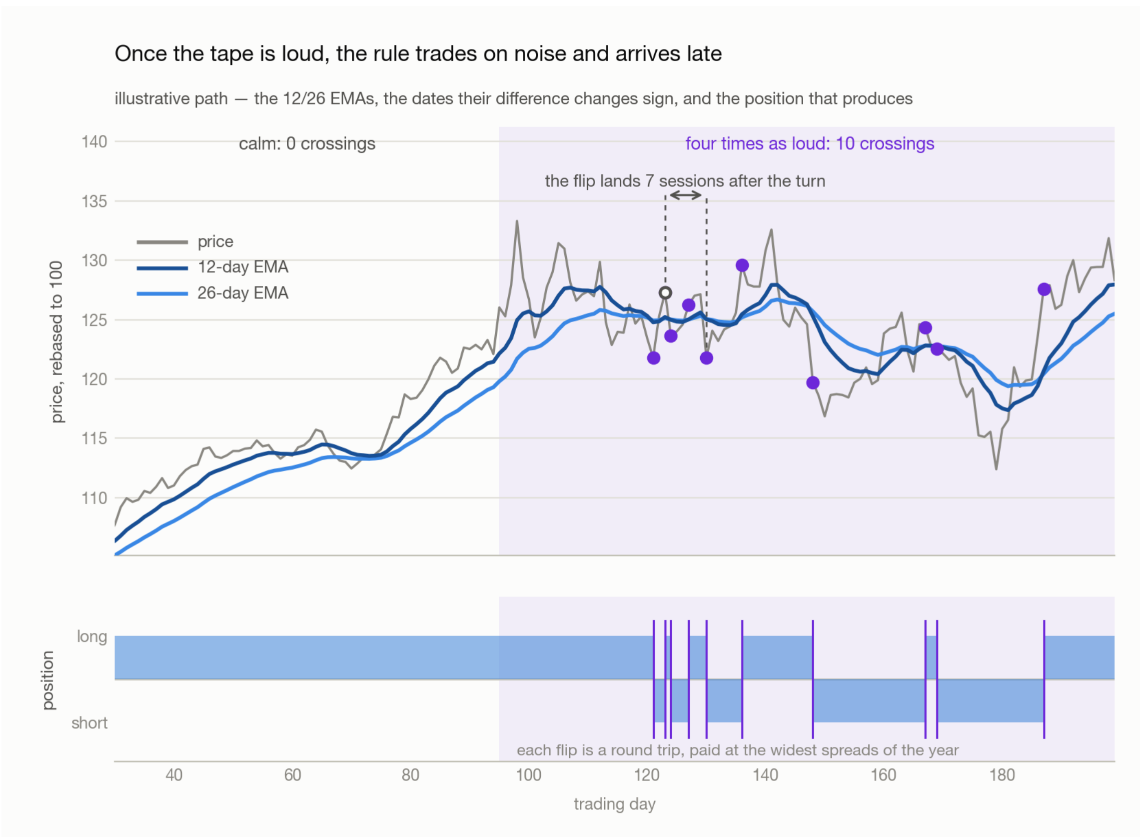


Figure 4.1: Above, one price path with a 12-day and a 26-day EMA over it. Through the calm first stretch the fast average sits clear above the slow one and the two never cross; through the shaded stretch, which has no net trend and four times the daily amplitude, they stay tangled and their difference changes sign ten times. Below, the position the rule holds: one unbroken long, then a shredded alternation of long and short blocks. Each flip is a round trip, paid at the widest spreads of the year. Illustrative.

One path, one rule, two regimes. In the calm stretch the fast average pulls clear of the slow one and stays there: no crossings, one position, held. In the shaded stretch the trend is gone and the swings are four times wider, so the two legs stay tangled and noise carries one across the other again and again.

Every flip also lands after the turn it is chasing, since an average only confirms a move that has already happened—so the rule goes short at the bottom of the spike and long at the top of the snap-back. Whipsaw is not scatter around the right answer; it is the opposite of it, repeated.

Standardizing the signal (§2.4) rescales it without moving a crossing date, and §3.4's smoother and deadband damp every date alike because neither knows the tape is violent. Clearing that ceiling means promoting the state of the market to a **variable**.

4.2 What the option market already knows

4.2.1 Volatility cannot be computed from the price

There is no way to read today's volatility off the price series. Every estimate built from past returns—a rolling standard deviation over a month, an EWMA, any weighting you like—is an average of what has already happened, so it only climbs as the new regime's returns accumulate inside its window. Shortening the window trades the lag for noise and buys nothing.

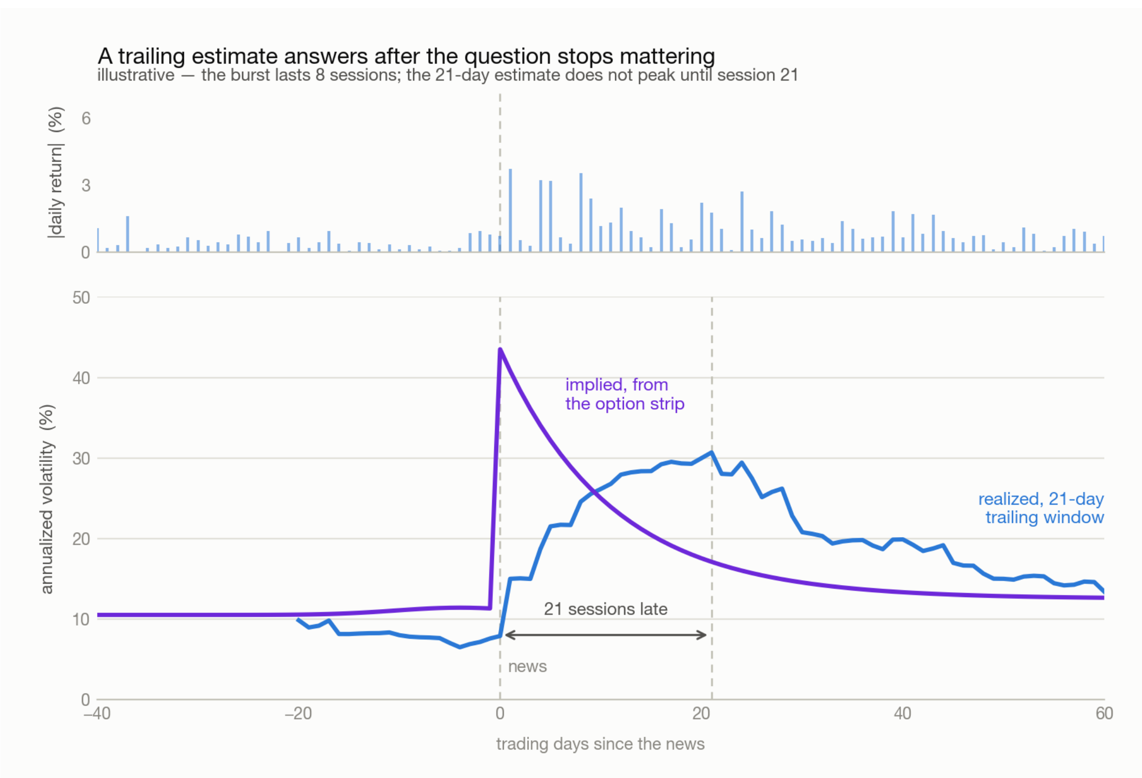


Figure 4.2: Above, the absolute daily return around a news event: quiet either side, with a burst of three and four percent moves in the sessions after it prints. Below, two answers to that event in annualized volatility. The implied line jumps on the day the news prints and decays smoothly back; the trailing 21-day line barely moves on the day and does not peak until session twenty-one, by which time the burst has been over for a fortnight. Illustrative.

The upper panel is the event and the lower panel is two answers to it. The trailing line does not peak until three weeks after the burst it is describing. **The other line is not an estimate at all**—it is a price, quoted by people who have to take a view on the next thirty days, and it moves the session the news does. Where it comes from is §4.2.2.

4.2.2 How an option hands you a volatility

An option fixes a price today for a trade that may happen later, and **the buyer chooses whether it happens**. Two of them, defined by which side that choice sits on:

	Call	**Put**
The holder may	**buy** the underlying at K	**sell** the underlying at K
Worth exercising when	$S_T > K$	$S_T < K$
Payoff at expiry	$\max(S_T - K, 0)$	$\max(K - S_T, 0)$
Pays off when the underlying	rises	falls
The most the buyer can lose	the premium	the premium
The most the buyer can make	unbounded	K, less the premium
Bought by someone who	wants upside without owning it	wants a floor under something they own

with K the **strike**—the price fixed in the contract—and S_T the underlying's price at expiry.

Both payoffs are **kinked**: flat on one side of the strike, one-for-one on the other. That kink is the whole instrument—it truncates the loss and leaves the gain open—and it is why an option is worth more when the underlying might travel further. Under Black–Scholes a call is worth

$$C = S\Phi(d_1) - Ke^{-R\tau}\Phi(d_2), \qquad d_1 = \frac{\ln(S/K) + (R + \sigma^2/2)\,\tau}{\sigma\tau^{1/2}}, \qquad d_2 = d_1 - \sigma\tau^{1/2}$$

where S is the spot price, K the strike, τ the time to expiry in years, R the risk-free rate, Φ the standard normal cumulative distribution function, and σ the volatility. Everything but σ is observable, which is what makes the equation worth inverting.

Definition 4.1 (Implied volatility)**.** The value of σ that makes the model reproduce the option's traded price—a forecast for the life of the contract, read out of a price rather than estimated from history.

Claim 4.2. The inversion is well defined: one price, one implied volatility.

Proof. The kink makes both payoffs convex in S_T. Raising σ spreads the distribution of S_T while leaving its forward mean where it was, and the expectation of a

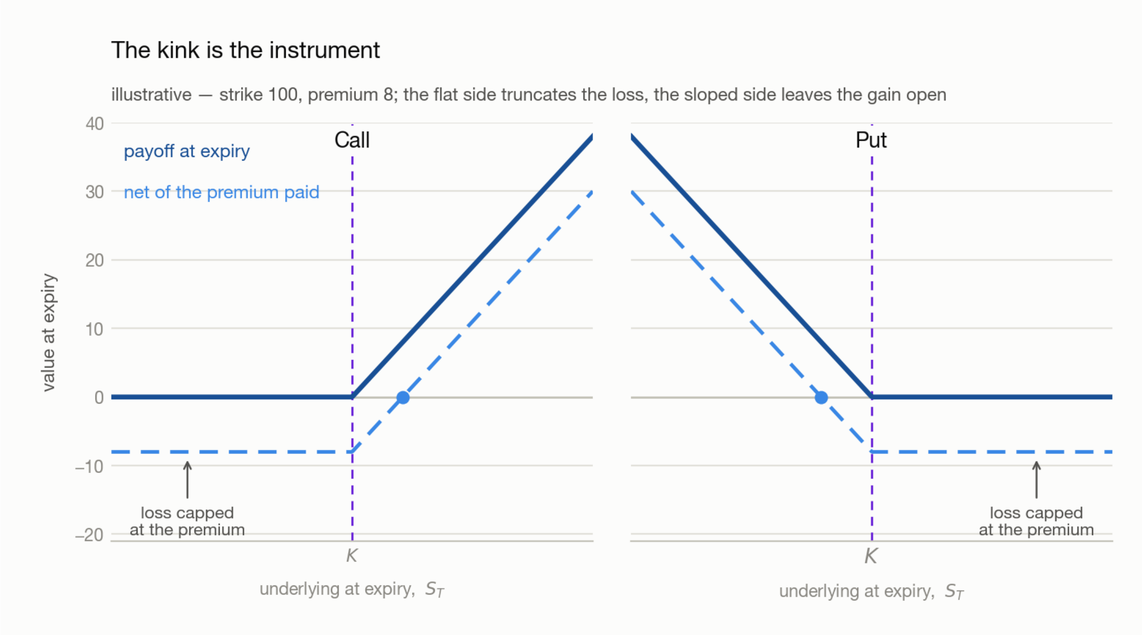


Figure 4.3: Value at expiry against the underlying's price at expiry, with the strike K marked in both panels. Left, a call: flat at zero below the strike, rising one-for-one above it, and a dashed line one premium lower for the same shape net of what was paid. Right, a put: the mirror image. Both dashed lines are flat on their far side—the loss is capped at the premium. Illustrative.

convex function rises under a mean-preserving spread. The price is therefore strictly monotone in σ, and a monotone map is invertible. □

There is no closed form for the inverse, so σ is solved numerically—bisection needs about twenty steps to reach four decimals, and monotonicity is what guarantees it converges. At the money the whole thing collapses to something you can do in your head:

$$C \approx 0.4\, S\sigma\tau^{1/2} \qquad \Longrightarrow \qquad \sigma \approx \frac{C}{0.4\, S\tau^{1/2}}$$

Example 4.3. A 30-day at-the-money call trades at 2 percent of spot. Quote the price as a fraction of spot and S cancels, leaving a denominator of $0.4\tau^{1/2} \approx 0.115$ at $\tau = 30/365$, so $\sigma \approx 0.02/0.115 \approx 0.17$—the market is charging for a **17 percent annualized** move over the next month. Read the same contract a day later at 3 percent of spot and the forecast is 26 percent: the number moves with the quote, which is the entire point.

Note (One contract is not the market)**.** That 17 percent is the volatility charged **at that strike**. Quotes on other strikes imply different numbers—the smile—so a single contract's answer is a property of one contract. §4.2.3 reads the whole strip at once to get rid of that dependence.

4.2.3 VIX

Definition 4.4 (VIX)**.** A constant-30-day implied volatility for the S&P 500, built from the whole strip of out-of-the-money options rather than from any one of them. For a single expiry the strip gives the implied **variance** directly, with no model to invert:

$$\sigma_\tau^2 = \frac{2e^{R\tau}}{\tau} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i) - \frac{1}{\tau}\left(\frac{F}{K_0} - 1\right)^2$$

where $Q(K_i)$ is the mid price of the out-of-the-money option struck at K_i, ΔK_i the gap between neighbouring strikes, F the forward level implied by put-call parity, and K_0 the first strike below F. Every term is a traded price, so the sum is the market's own variance rather than one contract's.

Two steps turn that into the published number. **Take the two expiries with τ between 23 and 37 days**—listed contracts do not fall on a 30-day schedule, and the band is what guarantees one on each side of the target—and **interpolate their variances linearly in τ** onto exactly 30 days. Then annualize, take the root, and quote it in percentage points:

$$\mathrm{VIX}_t = 100\left(\sigma_{30,t}^2\right)^{1/2}$$

Note (Decide whether your formula wants the variance or the volatility)**.** The published level is already a volatility—a VIX of 20 means an annualized 20 percent—because the strip computes a variance and the last step takes its root. Variance is what is additive, across time and across independent sources of risk; volatility is not. Anything that rescales a horizon or adds contributions needs $(\mathrm{VIX}_t/100)^2$; anything compared against a return in the same units needs the level. The error is invisible when made: a missing square root still leaves a plausible positive number.

Note (What the forecast is worth is a separate question)**.** Implied volatility is what the market charges, not what will happen—insurance is sold above its expected cost, and that gap is a strategy in its own right. Here the index is only a **state label**, for which a live consensus is enough.

4.2.4 One index for equities, another for rates

Claim 4.5. One equity index's implied volatility labels most sleeves, even though it is computed from one market.

The reason is what a panic does to correlations. In calm markets the sleeves move mostly apart; in a large enough shock the same levered participants sell all of them at once to raise cash and the cross-asset correlations run toward one. The states in which the label matters most are exactly the states in which one market's fear is every market's fear—an empirical regularity rather than a theorem, and better at the extremes than in the middle.

Sleeve	Regime index	What its spikes are about
Equities, credit, most commodities	**VIX**	growth and earnings shocks, and forced deleveraging
Rates, and anything priced off the curve	**MOVE**	policy surprises, inflation prints, auction stress

Note (Rates are the genuine exception)**.** Treasury volatility is close to unrelated to equity volatility, because the two are frightened by different things. A rate sleeve labelled by VIX is labelled wrongly, and **MOVE** is the same construction run on Treasury options. Neither index is in `CTA_data/` (100 · The Dataset), so both have to be fetched before §4.3 can be run.

4.2.5 From a level to a label

Definition 4.6 (Volatility regime)**.** A stretch of dates over which the amplitude of returns is roughly constant and materially different from the stretch on either side; a **regime shift** is the boundary between two of them.

§4.3 needs a small set of such states, named, because a continuous variable cannot be a row of a table. Write v_t for the index level on date t; its distribution is the obstacle.

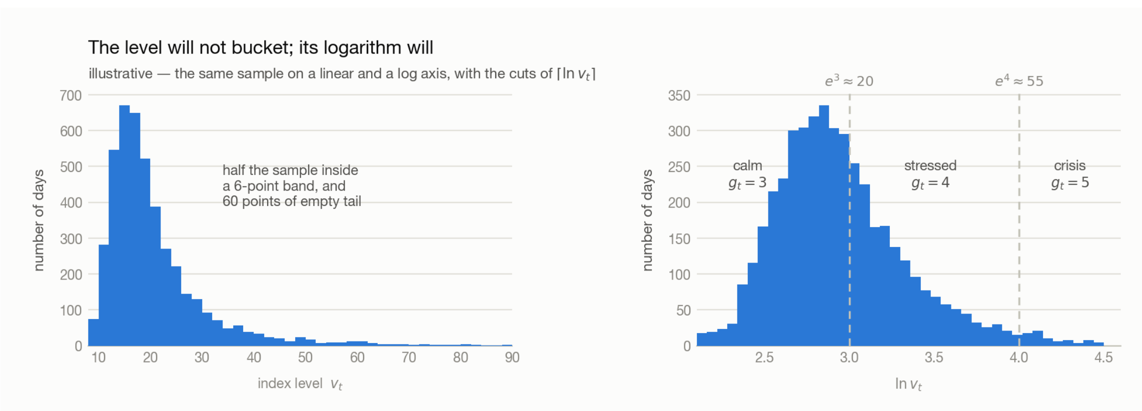


Figure 4.4: The same illustrative sample of volatility index levels, twice. Left, against the raw level: the mass piles into a narrow band and trails off into a long near-empty tail, half the sample sitting inside six points. Right, against the natural logarithm of the same values: close to symmetric, and the two integer cuts at 3 and 4—about 20 and about 55 on the level—divide it into calm, stressed and crisis.

The level spends most of its life in a band a few points wide and occasionally reaches four times that, so equal-width bins put nearly everything in the first one—the opposite of what §2.3.2.2's bucket test needs.

Definition 4.7 (Log-ceiling bucket)**.** Take the natural log of the level and round up to the next integer, $g_t = \lceil \ln v_t \rceil$, where g_t is the regime label on that date. Because ln compresses the sparse upper tail and stretches the crowded lower band, the integer cuts land at multiplicatively spaced levels:

Label g_t	**Level range**	**Reads as**
3	$v_t \in (7.4, 20.1]$	calm—the ordinary tape
4	$v_t \in (20.1, 54.6]$	stressed
5	$v_t \in (54.6, 148.4]$	crisis

The boundaries are e^3 and e^4, close to where a practitioner would have drawn them by hand. That is the construction's virtue: **the cuts use no data, so a date's label means the same thing in 2015 and in 2020.** Writing $g_t = \lceil c \ln v_t \rceil$ puts the boundaries at $e^{k/c}$ and recovers as many buckets as wanted; c is a hyperparameter like any other, and choosing it by which value produces the best-looking grid is what 5 · Modelling is about.

The alternative is §2.3.2.1's own machinery applied to the index: rank the dates by v_t and cut at the quintiles.

	Log-ceiling	**Rolling quantile**
Boundaries	fixed, the same in every year	move with the window
Group sizes	very unequal—crisis may be 3 percent of the sample	equal by construction
A label means	an absolute level of market fear	a rank against the recent past
Comparable across time	yes	no—the same v_t can be G5 one year and G3 the next
Look-ahead risk	none, the cuts use no data	real, if the window is not trailing

Note (The ranking must be rolling, never full-sample)**.** A quintile computed over the whole history uses 2020's levels to label 2015's dates, which is look-ahead bias of the plainest kind (§2.6)—and it is easy to commit here, because the index feels like context rather than like data.

4.3 Using the label

Each date now carries three numbers: a signal value, a forward return, and a regime label g_t. Two different questions can be asked of that—whether the edge **differs** by regime (§4.3.1), and whether the regime is **new information** the signals do not already carry (§4.3.2).

4.3.1 Conditioning — the two-way sort

Put the signal buckets across and the regime labels up, and report the mean forward return in every cell. It is §2.3.2's bar plot run once per regime—equivalently a **conditional model**: sort on the state first, then on the signal within it.

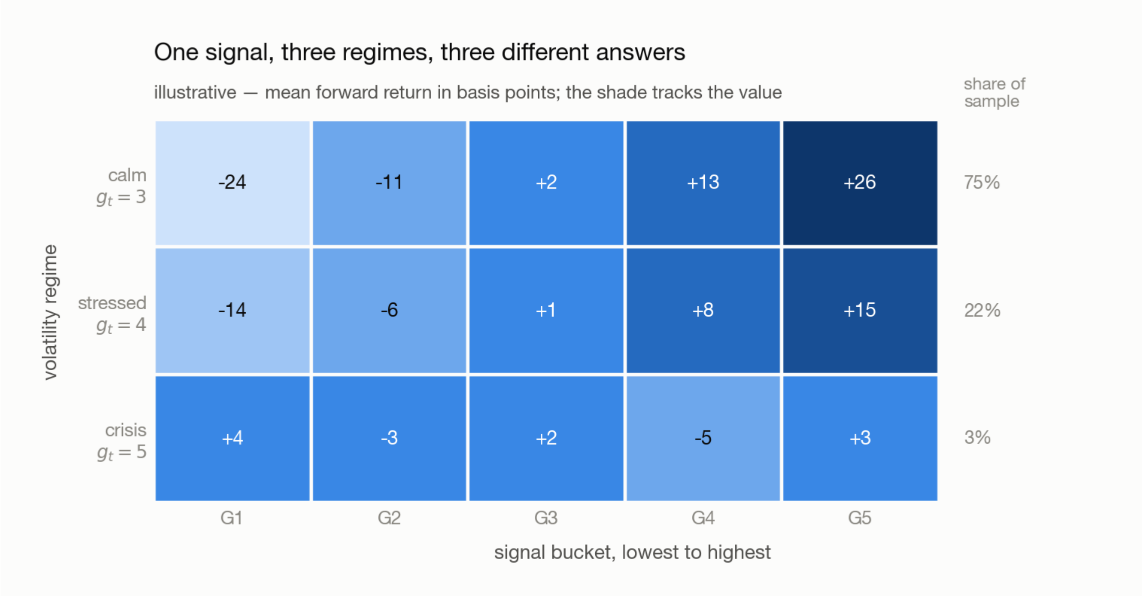


Figure 4.5: Mean forward return in basis points, signal buckets G1 to G5 across and volatility regime up: calm, stressed, crisis. The calm row, holding three-quarters of the sample, rises monotonically; the stressed row rises about half as steeply; the crisis row, 3 percent of the sample, has no ordering at all and its shading is flat across the row. Illustrative.

Read it by rows, and the reading is the finding. A staircase in the calm row that flattens through the stressed row and disappears in the crisis row is §4.1's picture measured on your own data: same rule, same parameters, less and less to show for them. The one-dimensional plot of 2 is the sample-weighted average of the three rows—it reports the calm row lightly diluted and never mentions that the other two exist.

Note (Keep the regime axis coarse, and print the counts)**.** The cells partition the same sample, so a one-way bar's $N/5$ dates become $N/15$ across three rows and every error bar (§2.3.2.3) widens by $3^{1/2} \approx 1.7$—at five rows, $5^{1/2} \approx 2.2$. Worse, the row you most want is the thinnest: crisis cells hold roughly $N/167$ dates, error bars about 5.8 times wider, so **a flat crisis row is equally consistent with an edge nobody has enough crisis days to measure.** Report the count beside every mean, and treat a row you cannot measure as unknown rather than as zero.

4.3.2 Residualizing — is any of it new?

Regress the candidate on the incumbents and keep what is left:

$$x_t = \beta_0 + \sum_j \beta_j x_{j,t} + \epsilon_t$$

where x_t is the candidate signal on date t, $x_{j,t}$ the signals already in the book — momentum and MACD, here—β_j their fitted coefficients, and ϵ_t the residual. Then run §2.3.2's bar plot on ϵ_t in place of x_t.

Claim 4.8. If the staircase survives on ϵ_t, the candidate carries an edge the existing signals do not already own.

Proof. Least squares sets the derivative of the summed squared error to zero, giving one normal equation per regressor—$\sum_t \epsilon_t x_{j,t} = 0$ for every j, and $\sum_t \epsilon_t = 0$ from the intercept. Zero sum of products on a centred variable is zero sample covariance, so $\rho(\epsilon, x_j) = 0$ for every incumbent. By §2.1.2 a portfolio's expected return runs entirely through the correlation between what sets the weights and the forward return, so weights built on ϵ_t earn whatever they earn somewhere the book is not standing. □

Note (Orthogonal is not independent)**.** Zero *linear* correlation is all least squares buys. A residual that is large exactly when MACD is large **in absolute value** is a function of MACD the regression cannot see—and that is not a corner case here, since the candidate is a variable about σ and |MACD| widens with σ too. Add the transform you suspect as an extra regressor and residualize against that too.

Note (With fifty signals, regress against the model's output)**.** Fitting a candidate against fifty regressors on overlapping daily data fits noise long before it exhausts the information, and fifty pairwise regressions give answers that do not compose. The book already produces one number per asset per date—its combined prediction p_t. Use that single series as the regressor: one residual, and the question it answers is the one actually being asked.

4.3.3 Where the answer is spent

Often not in the return model at all. The natural home for §4.3.1's grid is the **optimizer**—a penalty that rises with the regime, an exposure cap that binds in the crisis row, a volatility target that scales the whole book—which is why the grid earns its place even when §4.3.2's residual comes back flat. Answering *yes* to §4.3.1 and *no* to §4.3.2 is a common and perfectly good result, and performance evaluation is where it is spent.

Add one dimension per experiment, and record what it bought—what the step adds alone, whether it overlaps what is already there (§4.3.2's residual), and which direction it is useful in, since a variable can predict return poorly and volatility well. The record has to pass interpretability in an operational sense: move an input and you should be able to say in advance, roughly, what the output does. Measuring 8 bp where you expected 10 is fine; *"it moved and I do not know by how much"* is not, and it is what adding several things at once guarantees.

Note (The scorecard is regime-dependent too)**.** Every chart above reports mean forward return with an error bar, for the reasons §2.5 sets out. A Sharpe ratio has the tape's volatility in its own denominator, so the same number means different things in different rows: 0.8 earned in a year when the index itself ran at 0.4 is a far harder result than 0.9 earned in a year when buying the index and doing nothing paid 0.8. Collapsing the regime axis into one ratio destroys exactly the context needed to read the ratio.

Background

Why is an option the instrument a volatility forecast comes from?

- **Its value depends on the range, not the direction.** The flat half of the payoff truncates the losses, so a wider distribution of outcomes is worth strictly more—which makes an option the most volatility-sensitive thing quoted, and §4.2.2's inversion stable enough to be worth doing.
- **Out-of-the-money contracts are close to pure volatility bets.** One struck far above the price pays only if the underlying travels an unusual distance, so its price is almost entirely a statement about the tails—which is why VIX reads the whole strip, not the at-the-money contract.
- **The forecast has a fixed horizon attached.** Every contract has an expiry, so an implied volatility is always *volatility over the next* τ *days*—which is what allows two of them to be interpolated onto a constant 30-day horizon.

Appendix · Notation

Throughout, t is the date. The regime index is a property of the market rather than of one asset, so it carries no asset subscript s.

Symbol	Means	First used
σ	the volatility of the underlying—the amplitude of its move, without regard to sign	§4.2.2
C, S, K, τ, R	a call's price, the spot price, the strike, the time to expiry in years, and the risk-free rate	§4.2.2
Φ, d_1, d_2	the standard normal cumulative distribution function and the two arguments Black–Scholes feeds it	§4.2.2
S_T	the underlying's price at expiry	§4.2.2
$Q(K_i)$, ΔK_i, F, K_0	one out-of-the-money option's mid price, the gap between neighbouring strikes, the forward level, and the first strike below it	§4.2.3
σ^2_τ, $\sigma^2_{30,t}$	the implied variance the strip gives for one expiry, and the same interpolated onto 30 days	§4.2.3
v_t, g_t, c	the index level on that date, the regime label cut from it, and the multiplier setting how fine the cuts are	§4.2.5
x_t, $x_{j,t}$, β_j, ϵ_t	the candidate signal, the signals already in the book, their fitted coefficients, and the residual	§4.3.2

Symbol	Means	First used
p_t, N	the model's combined prediction, and the number of dates in the sample	§4.3

Note (Collisions to watch)**.** σ is a market-wide volatility here where §2.4's $\sigma_{s,t}$ is one asset's trailing estimate, and R is the risk-free rate discounting an option where §2.5's r_f is the same rate subtracted from a portfolio return. Uppercase S is a spot price and lowercase s is 2's asset index; Φ is a normal CDF and N a sample size, which is why the CDF is written Φ rather than N. Lowercase g_t is a regime label on the vertical axis while §2.3.2's uppercase G1 . . . G5 are signal buckets on the horizontal one, and §4.3.1's grid has both at once. ϵ_t here is a signal residual after regressing on other **signals**; §2.2's ϵ is the part of the forward **return** no signal reaches.

5 · Modelling

5.1 Why one split is not a backtest

5.1.1 The textbook split

Definition 5.1 (Split)**.** A *split* cuts the sample by date into three contiguous blocks—**train**, **validation** and **test**—each of which is permitted to do exactly one thing to the model.

Block	**What it is for**
train	Where the coefficients are estimated. It is the only block that may move a coefficient, and the only dates the fit ever sees.
validation	Where the choice *between* fitted models is made—which features survive, which λ, which model. It scores; it never fits.
test	Scored once, at the end, and it chooses nothing. The moment any decision is taken on it, it stops being out of sample.

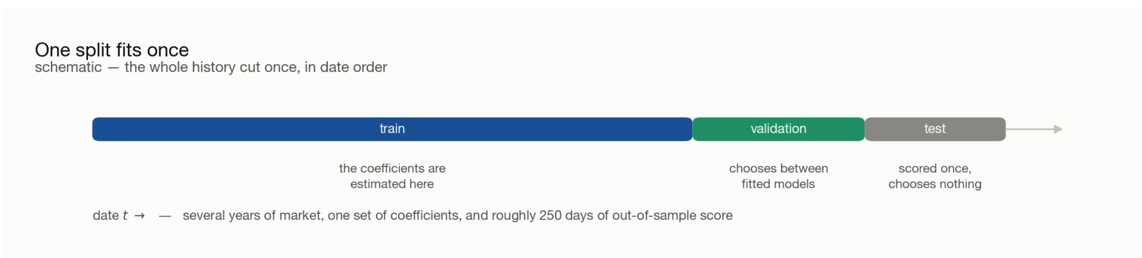


Figure 5.1: The textbook pipeline: one history, cut once, in date order. Illustrative proportions—train carries several years, validation and test roughly a year each.

The order is not a convention. The blocks run forward in time because a model fitted on next year and scored on last year has learned something no live book could have known.

5.1.2 Why the single split fails

It produces a number. It also has three defects, and they compound.

Defect	What goes wrong
One fit discards time	One set of coefficients over several years asserts that those years were one market. Set early-pandemic oil—no demand, and a consensus that it would never return—against the war-driven energy rally that followed: a coefficient fitted across both describes neither.
Almost nothing is left to score	A year of validation is roughly 250 dates, holding whichever regimes happened to fall inside it. The split never asks what the strategy does when the regime turns, and a drawdown measured over one calm year is not a risk estimate.
The sample is small and the signal is faint	In finance **0.1 to 0.2 is already a good** R^2 (9), against 0.5 for a school dataset and 0.9 in physics. Weak dependence with few observations is exactly where a fit lands on noise: ten points from a cloud reach R^2 0.8 easily.

5.2 The walk-forward ladder

The three defects of §5.1.2 share one cause: the model is fitted once. So fit it many times. Cut the training history into short segments, put the same train–validation–test frame on the first few of them, and slide that frame forward one segment per iteration until it reaches the end of the training set. Each rung is a complete train–validation–test cycle in miniature, and there are as many of them as the history allows.

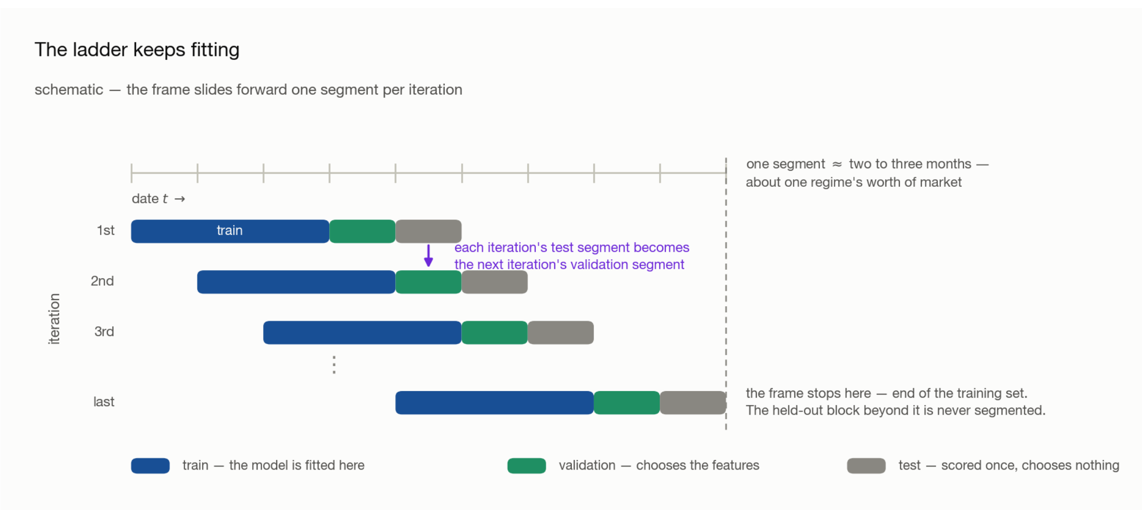


Figure 5.2: The ladder that replaces the single split: the frame advances one segment per iteration, so each row's test segment is the next row's validation segment. Schematic—the ruler stops at the end of the training set, since the held-out block beyond it is never segmented.

Note. The ruler ends where the training set ends. What lies beyond it—the held-out block that never enters the ladder—is a different object with different powers; that is

§5.3.

5.2.1 How long a segment should be

Definition 5.2 (Segment)**.** A *segment* is a short contiguous block of dates—here **two to three months**—treated as one regime.

The length is set by how fast the market can change its mind, and the market is made of people. Within a day a regime shift is implausible: individuals change their view constantly, the crowd does not. Within a month, after a run of large news, one shift is plausible. Within a year or two there is room for several, because that is enough time for the same participants to change their minds repeatedly. So a segment must be short enough that one segment is approximately one regime, and long enough to hold a usable sample.

5.2.2 The slide

Number the segments $1, 2, 3, \ldots$ and the frame moves like this:

Iteration	Train	Validation	Test
1	1–3	4	5
2	2–4	5	6
3	3–5	6	7

Claim 5.3. The ladder repairs all three defects of §5.1.2 at once.

Each fit now sees one regime's worth of data and is re-estimated as the market moves on, so time is no longer averaged away. Every segment past the first frame is eventually scored out of sample, so the out-of-sample record grows with the length of the history instead of being fixed at one year. And a regime shift that lands inside a validation segment is now a *test* the strategy has to pass, rather than an event the single split happened to miss.

Note. Read the table down its two right columns: **each iteration's test segment is the next iteration's validation segment.** Nothing is discarded between rungs, and no segment is ever scored twice under the same coefficients.

5.2.3 One rung, in full

A rung is not a single fit. Train ends, a validation segment intervenes, and only then does test begin—so a model fitted on train alone is asked to predict across a gap of market it has never seen, and pays for it.

The repair is the two-step shape that Lasso already imposes. Lasso adds an L_1 penalty to the least-squares objective,

$$\min_{\beta} \quad \sum_t \left(y_t - \sum_j \beta_j x_{j,t} \right)^2 + \lambda \sum_j |\beta_j|$$

where y_t is the forward return on date t, $x_{j,t}$ signal j's value on that date, β_j its coefficient, and $\lambda \geq 0$ the penalty strength. The penalty drives coefficients exactly to zero, which makes it a **feature selector**; and because the surviving coefficients are biased toward zero by that same penalty, the standard practice is to **refit** the survivors without the penalty before predicting.

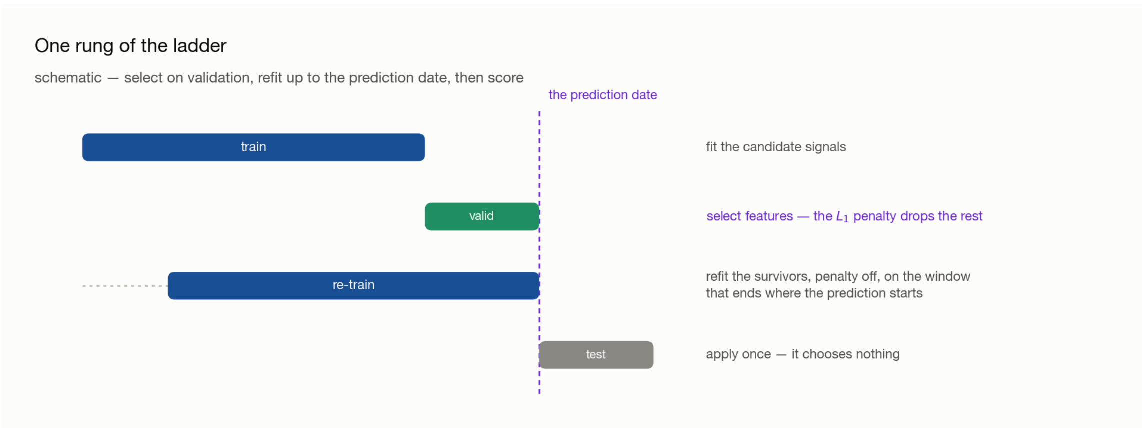


Figure 5.3: One iteration, in four stages: fit on train; select features on the validation segment; refit the survivors on a window shifted forward so that it ends where the prediction starts; apply once to test, beyond the dashed prediction date.

So each rung runs: **fit on train**, **select features on the validation segment**, **refit the survivors on a window ending where the prediction starts**, **apply once to test**.

Note (Two house styles)**.** The refit window is either the training window shifted forward, or the training window plus the validation segment. The sample sizes differ little and both are used in industry; it is a preference, not a correctness question. What is not optional is that the window ends at the prediction date—see §5.4.3.

5.3 Two validation layers

The validation *segment* inside each rung and the large held-out validation *block* at the end of the history are different objects with different powers.

Figure 5.4: The two layers. The inner layer is the miniature ladder inside the training block, which chooses which features survive; the outer layer is the ringed held-out validation block, which chooses the model class, its hyperparameters, and the optimizer's parameters.

Layer	May choose	May not
Inner—the validation segment inside each rung	which **features** survive, and the refit that follows	anything held constant across rungs
Outer—the held-out validation block	the **model class**, its **hyperparameters** (λ), and the **optimizer's parameters**	coefficients—those are fitted on train
Test	nothing	everything

5.3.1 Why the inner layer cannot be skipped

Exploration already looked at which signals worked *in the training period*. Of course the fit then describes that period well: momentum was selected partly because it was seen to work there. That says nothing about whether it was useful going in, and the only way to find out is to select on dates the selection did not see—which is what the validation segment is for.

Note. A signal's usefulness is not a constant either. When more signals enter the book, or when one starts to fail, momentum should no longer carry the weight it carried alone. Because the ladder re-selects at every rung, that adjustment happens on its own rather than being asserted once.

5.3.2 What only the outer block may decide

λ is not fitted, it is searched: run the ladder once per value on a grid—0.1, 0.5, 0.7, 1—and compare. The same is true of the model class itself (Lasso, Ridge, a decision tree). Both comparisons need dates that no rung of the ladder consumed, which is exactly what the held-out block is.

5.3.3 The optimizer's parameters

The prediction is not yet a portfolio. Weights come from a mean–variance problem: choose w to

$$\max_{w} \quad E[r_p] - \gamma \cdot \mathrm{Var}[r_p]$$

subject to constraints such as

$$\sum_i w_i = 1.0 \qquad \sum_i |w_i| \le 2.0 \qquad |w_i| \le 0.20$$

—net exposure 100 percent, gross exposure at most 200 percent, and no single asset above 20 percent—where w_i is the weight on asset i, $E[r_p]$ and $\mathrm{Var}[r_p]$ the portfolio's expected return and variance, and $\gamma \ge 0$ the **risk-aversion coefficient**.

Two of these are free parameters, and neither can be reasoned to:

- **γ.** Nobody can say from first principles whether a risk appetite is 0.3, 0.5 or 1. The number has no interpretable units—it is a trade rate between two quantities measured in different things—so it is chosen by running the grid and reading the result.
- **The per-asset cap.** With 37 assets, a 20 percent cap may be too tight for the book to hold enough of whichever names are paying; 25 or 30 percent may be the right ceiling. Gross and net exposure, by contrast, are set by mandate and are not tuned.

Note. All of this tuning belongs in the **outer** block. Tuning γ inside the rungs would let a portfolio parameter be chosen on the same dates the features were chosen on, and the held-out block would then be measuring a strategy that had already been fitted to it.

5.4 Preparing a signal for a model

Two transformations stand between a raw signal and a model, and both exist to stop the fit from tracking things that are not the market.

5.4.1 Clipping: keep the observation, cap the value

Claim 5.4. A ranked signal does not fill its range evenly—the mass piles at both ends—and those ends move the fit far more than their count suggests.

The first half is empirical: momentum in particular pins at an extreme whenever a name runs, so the histogram is a barbell rather than a plateau. The second half is mechanical. In a least-squares fit each observation's pull on the slope grows with its distance from the mean of x, so a handful of far points can swing a coefficient that

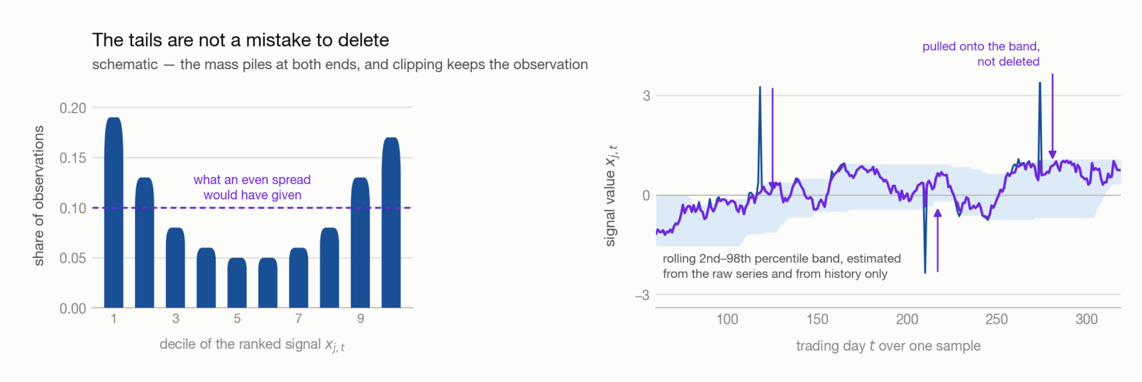


Figure 5.5: Left, the share of observations by decile of the ranked signal—high at the first and last deciles, low in the middle, against the dashed line at 0.1 that an even spread would give. Right, a signal series against a shaded rolling percentile band, with the spikes that reach outside it pulled back onto its edge.

hundreds of central points barely move. The coefficient then jumps for reasons that have nothing to do with the market.

Deleting the far points, which is the textbook move, fails twice here:

- **The sample is already small.** Every deletion sharpens §5.1.2's third defect, and in a messy sample it is not even clear which points are outliers.
- **The tail carries information.** What the book does when a signal goes extreme is exactly what the model should learn. Delete those dates and it never can.

Definition 5.5 (Winsorization)**.** Replacing any value outside a chosen quantile band by the nearer edge of the band:

$$x_{j,t}^{\text{clip}} = \min\left(\max\left(x_{j,t},\ Q_{j,t}^{\text{lo}}\right),\ Q_{j,t}^{\text{hi}}\right)$$

where $Q_{j,t}^{\text{lo}}$ and $Q_{j,t}^{\text{hi}}$ are quantiles of signal j over the trailing L dates—1 and 99 percent, or 2.5 and 97.5 percent; there is no canonical pair.

Note (Compare raw to raw)**.** The band for date t must be estimated from the **unclipped** history. Feed yesterday's capped value into today's quantile and the band walks inward on itself, a little further every step. Keep the raw column and produce the clipped one beside it.

Note (Why the band rolls)**.** A value that is extreme against the last three months may be unremarkable six months later, once the market has moved to it. A fixed threshold pins such a signal at the 99th percentile forever; a rolling one releases it as soon as the surrounding distribution catches up.

→ `rolling`, `quantile` and `clip` as pandas calls: 8 · Toolbox.

5.4.2 Standardizing: put every signal on one scale

Momentum lives in roughly $[-0.05, 0.05]$. VIX lives in $[0, 100]$. Fit both and the coefficients absorb the mismatch.

Claim 5.6. Coefficient magnitudes carry no information about a signal's importance until the signals share a scale.

Proof. Replace x_j by $c\,x_j$ for some $c > 0$. The fitted values are unchanged if β_j is replaced by β_j/c, and least squares chooses fitted values—so the fit is identical and the coefficient is whatever the units make it. A comparison of β magnitudes across differently scaled signals is therefore a comparison of units. □

The consequence is not merely aesthetic. Every input is an **estimate** and carries error: a momentum measured from returns is the residue of participants entering and leaving, and a jump in it is as likely to be noise as news. A large β multiplies that noise into the prediction, while the small β on the wide-ranging signal cannot respond when it genuinely moves.

Definition 5.7 (Standardized signal)**.**

$$z_{j,t} = \frac{x_{j,t} - \mu_j}{\sigma_j}$$

with μ_j and σ_j the mean and standard deviation of signal j over the estimation window. Dividing by a rolling standard deviation, or by a cross-sectional sum, are alternatives; what matters is that **every signal is treated the same way**.

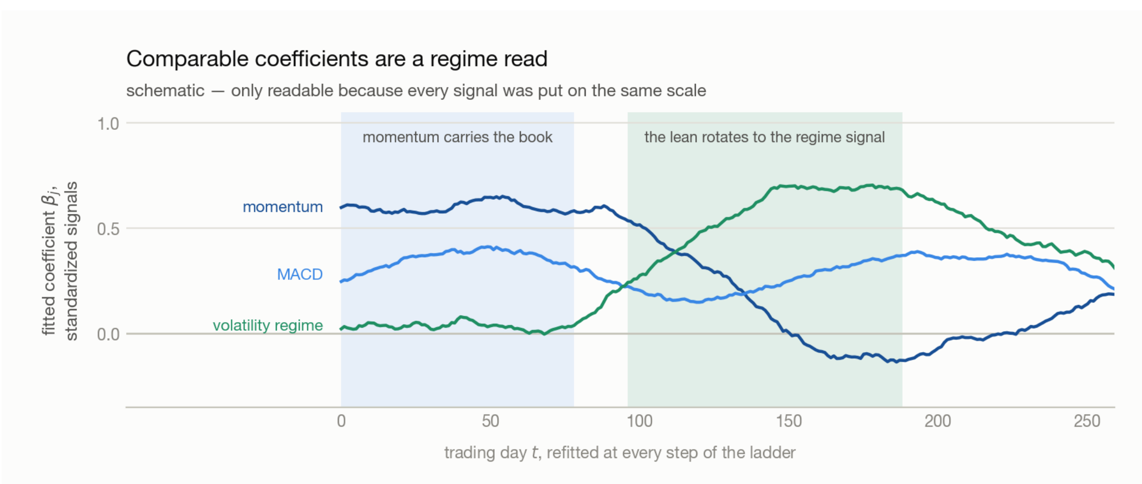


Figure 5.6: Three coefficient paths over a year of refits: momentum decays from high toward zero, a volatility-regime signal rises from zero to become the largest, and MACD stays between them. The shaded stretches mark where momentum carries the book and where the lean rotates to the regime signal.

Three things follow, in increasing order of usefulness:

Consequence	**Why**
Coefficients become comparable	With units removed, β_j measures response per standard deviation of signal—the same question for every signal.
The coefficient path becomes a regime read	Plot every β_j from every rung on one axis. Momentum dominating one stretch and volatility another **is** the change in market sentiment, in a form you can look at.
Errors partly cancel	On one scale, with comparable coefficients and errors that are close to independent across signals, the noise in the combined prediction averages down as signals are added, instead of being dominated by whichever signal has the largest units.

5.4.3 Which mean and standard deviation you may use

μ_j, σ_j and the quantile band all have to be estimated from somewhere, and that choice is where look-ahead enters.

The rule follows from **where the model is built**: at the *end* of the training window. Every date inside that window is history at that moment, so standardizing with the whole training window's mean and standard deviation is legitimate—even though, from the point of view of an individual date early in the window, the statistic used to scale it was computed partly from its own future.

At this point of construction	**These statistics are available**
End of the training window	the whole training window
End of the training window	**never** the validation segment, and never test
End of a refit window	the whole refit sample, validation segment included

Note. The distinction is between *a signal's* information set and *the model's*. A signal computed on date t may use only data up to t, because it has to be computable live. The model is not computed on date t; it is computed once, at the end of its window, and then applied forward. Confusing the two costs either a leak or an estimate worse than the one you were entitled to.

Note. This is the failure worth being paranoid about. In a time series the smallest leak makes the result beautiful, and the gap between a beautiful backtest and the live book is where the money goes.

Appendix · Notation

Throughout, t is the date, i indexes assets and j indexes signals.

Symbol	Means	First used
$x_{j,t}$, $x_{j,t}^{\text{clip}}$, $z_{j,t}$	signal j on date t: raw, after clipping, after standardizing	§5.2.3, §5.4.1, §5.4.2
y_t, β_j	the forward return being predicted, and the coefficient the fit puts on signal j	§5.2.3
λ	the L_1 penalty strength—searched on a grid, not fitted	§5.2.3
L	the trailing window, in dates, that the clip band is estimated over	§5.4.1
$Q_{j,t}^{\text{lo}}$, $Q_{j,t}^{\text{hi}}$	the low and high quantiles of that window	§5.4.1
μ_j, σ_j	the mean and standard deviation used to standardize signal j	§5.4.2
w_i, γ, $E[r_p]$	the portfolio weight on asset i, the risk-aversion coefficient, and the portfolio's expected return	§5.3.3

Note (Collisions to watch)**.** σ_j here is one **signal's** dispersion, used only to rescale it—not §2.4's $\sigma_{s,t}$, an asset's trailing volatility, and not §4.1's σ, the amplitude of the tape. β_j is a coefficient on a **signal**, as in §4.3.2, never a market beta. λ is the penalty and γ the risk aversion: the lecture wrote both as λ and then renamed the second, and they are unrelated. L is a trailing window as in §4.2.1, here over a signal rather than over returns.